Climate Change and the Macroeconomics of Bank Capital Regulation*

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Abstract

This paper studies the interaction between bank capital regulation and climate policy through the lens of a quantitative multi-sector DSGE model with bank failure and firm default. We find that *tilted* bank capital requirements can support a re-allocation away from the fossil energy sector and reduce aggregate emissions, although the effect is small compared to emission taxes. Since tilted capital requirements also reduce liquidity provision to households, the associated welfare gains are even smaller than for modest emission taxes and can become negative. By contrast, *optimal* capital requirements decline across all sectors if emission taxes are raised, in order to counteract a secular decline in aggregate loan demand and liquidity supply, while they slightly increase in response to climate policy uncertainty.

Keywords: Bank Capital Regulation, Macro-Banking Model, Macro-Finance Model, Climate Policy, Multi-Sector E-DSGE Model

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1 Introduction

Limiting the emission of socially harmful greenhouse gases is one of the largest challenges for economic policy in the next decades. A meaningful emission reduction requires a shift away from fossil to clean energy sources. How economic policy should induce such a sectoral re-allocation and how it should account for the associated financial flows is subject to an active discussion by policymakers. This can be seen from the number of speeches by central bank officials that mention both ”climate change” and ”capital regulation” (Figure 1). Two distinct issues are discussed most prominently. First, the insufficiently low levels of emission taxes currently in place have sparked interest in financial regulation that directly supports the re-allocation away from the fossil sector, such as tilted bank capital requirements. It remains unclear whether such a policy can induce a quantitatively relevant sectoral re-allocation, in particular in comparison to emission taxes. Second, there is an active discussion about optimal bank capital regulation if emission taxes induce the required sectoral re-allocation, how bank regulation should deal with climate policy uncertainty, and whether bank regulation can facilitate more ambitious climate policy by softening the adverse short run consequences of high emission taxes.

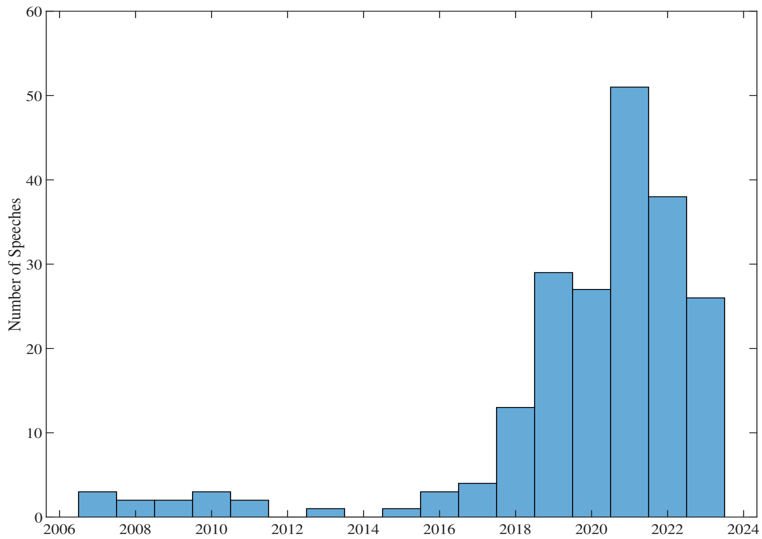

Figure 1: Climate Change and Capital Requirements. The number of Central Bank speeches mentioning ”Climate Change” and ”Capital Requirements” is computed using the database provided by Campiglio et al. (2025), see Appendix C for details.

This paper answers these two distinct sets of questions through the lens of a novel quantitative macroeconomic framework. Our model combines elements from the macro-banking and macro-corporate finance literature with a multi-sector production structure that is typically used in the assessment of climate policies. Non-financial firms finance their investment with equity and defaultable loans from banks (Gomes et al., 2016). Banks are financed with deposits and equity, and we allow for the possibility of idiosyncratic bank failure following Clerc et al. (2015).[1] Households are protected from bank failure by a bailout guarantee, i.e. they are repaid using taxpayer funds if a bank fails. Hence deposits are nominally risk free and households are willing to pay a liquidity premium to hold them. Together with the bailout guarantee, liquidity premia imply that deposit financing is always cheaper than equity financing for banks. To ensure that they hold positive equity, the regulator enforces minimum capital requirements (Begenau, 2020).[2] Tighter bank regulation decreases the bank failure rate but simultaneously reduces the provision of liquid deposits to households (Van den Heuvel, 2008; DeAngelo and Stulz, 2015). The socially optimal bank capital requirement balances these effects.

We embed this tractable model of optimal bank capital requirements in a multi-sector production economy with nominal rigidities that can reconcile key properties of the energy sector, climate policy and business cycles. Final good producers combine intermediate inputs from three different sectors that consist of a representative clean energy, fossil energy, and non-energy firm. While both energy goods are highly (but not perfectly) substitutable, the elasticity of substitution between energy and non-energy goods is small (but not zero). Fossil energy firms generate emissions in the production process. The public sector taxes emissions and firms can reduce them by paying a cost, which can be interpreted as emission abatement (Heutel, 2012) or a switch to less profitable clean energy sources (Acemoglu et al., 2012).

Bank capital regulation is linked to the production side through the corporate loan market. The loan price offered by banks declines ceteris paribus in banks' cost of capital,

[1]Our model purposefully focuses on the failure of individual banks and abstracts from systemic financial stability concerns. This is motivated by recent empirical (Jung et al., 2024) and quantitative macro results (Kaldorf and Rottner, 2025), who find that the effects of ambitious climate policy on systemic risk are non-negligible but manageable using standard macroprudential instruments.

[2]Rationalizing bank capital requirements by a bailout put goes back at least to Kareken and Wallace (1978). Pennacchi (2006) demonstrates that deposit insurance is critical to bank liquidity provision but that it also creates moral hazard. The interplay between bailout guarantees and liquidity premia is studied in Gorton and Pennacchi (1990).

which in turn depends on the capital requirement and the relative benefit of deposit financing. The demand for loans is shaped by two credit frictions in the production sector. On one hand, firm managers are more impatient than banks, i.e. they prefer to front-load dividend payouts by taking up loans. On the other hand, firms are subject to uninsurable idiosyncratic productivity shocks and they default if current revenues fall short of the due loan repayment (Gomes et al., 2016). Default entails a restructuring cost for banks which is fully reflected in loan prices. Firm managers balance the cost of loan financing (restructuring cost) with its benefits (dividend front-loading).

The optimal investment decision depends on discounted expected revenues and takes into account that holding more capital reduces the *future* default probability and thereby increases *current* loan prices. This enables firms to increase current dividends. By creating debt capacity, investment has a dual purpose in our model, which is crucial for the transmission of (tilted) bank regulation to the real sector. Raising capital requirements lowers the loan prices offered by banks so that the associated benefits of holding capital decline and firms invest less.

We calibrate the model to match salient empirical features of financial markets, climate policy, and the sectoral structure of the economy. Using our calibration as a laboratory for different policy experiments, we first assess to which extent bank capital regulation can act as a climate policy instrument. Setting fossil penalizing capital requirements to their maximum of 100% raises firms' cost of debt by almost 100 basis points above other sectors, as the fossil loan rate does not include the funding advantage stemming from the liquidity premium on deposits and the bailout guarantee. In our baseline calibration, this induces a mere 1% decline of emissions in the long run, which is for three reasons. First, this increase in loan rates translates into a much smaller increase in the overall cost of capital as only around 30% of capital is financed by loans. Second, while creating debt capacity is less important for fossil firms if loan financing becomes more expensive, this effect is quantitatively small, as the marginal impact of investment on the default probability is rather low. Third, simple fossil penalizing capital requirements do not affect the share of abated emissions within the fossil sector. Conceptually, their effect on emissions resembles a (small) tax on fossil capital.

Therefore, we allow for sustainability linked capital requirements (SLC), which explicitly condition banks' fossil capital requirement on the abatement effort undertaken by fossil energy firms. Such a policy "engages" banks in the net zero transition, as it rewards emission reduction activities at the firm level. By contrast, the simple fossil penalizing capital requirement resembles a policy-induced "divestment" strategy. We assume that the SLC takes on the value of 100% if fossil energy firms do not abate, while they receive the baseline 8% capital requirement if they fully abate their emissions. Under this policy, emissions decline by around 4.5%, which turns out to be considerably smaller than the effects of the (coverage-weighted) average global emission tax of 5 US-dollar per ton of carbon dioxide ($/tCO2) that is currently in place.

The total welfare effect takes the deposit channel of bank capital regulation into account, as deposit supply declines sharply if banks are required to hold more equity for fossil loans. Tilted bank regulation faces a trade-off between supporting climate policy and reducing liquidity provision to households. The net welfare effect is -0.14% for fossil penalizing capital requirements but amounts to +0.38% for the SLC. Crucially, the small global emission tax that is currently in place implies much smaller adverse side effects on liquidity provision, such that tilted bank capital regulation is generally an inferior climate policy instrument.

We then turn to the case in which climate policy is implemented using emission taxes and study the implications for optimal bank capital regulation. How does the emission tax level affect the bank failure rate and the amount of liquid deposits? These variables are key in determining optimal capital requirements in our model. Consider a permanent increase in the tax level. Since clean, fossil, and non-energy goods are imperfect substitutes, aggregate investment and loan demand contract and banks reduce their balance sheets. The supply of deposits contracts and the deposit spread widens. This makes deposit financing cheaper for banks, such that they offer higher loan prices ceteris paribus. This improvement in firm financing conditions (partially) mitigates the negative investment and loan demand effects of higher taxes, but this comes at the cost of higher leverage ratios and default rates in the corporate sector. Importantly, default risk increases symmetrically in all sectors. By contrast, the bank failure rate is not directly affected by the tax level in the long run since it is essentially fixed by the capital requirement. As a consequence, the optimal capital requirement declines from 8% in the baseline without emission taxes to 7.4% for emission taxes consistent with net zero.

We then consider optimal bank capital regulation in the presence of climate policy uncertainty, which is a pervasive feature of environmental regulation. Different from the tax level, this bears direct implications for the average corporate default and bank failure rates, which both rise if emission taxes are subject to persistent shocks. The reason is that even though taxes vary symmetrically around their long run level (which can be interpreted as the trend), default risk is a non-linear function of the emission tax, i.e. it increases more if taxes suddenly rise than it declines if taxes suddenly drop. By contrast, deposit supply is approximately constant over time such that it is optimal to tighten bank capital requirements by around 0.1 percentage point.

Our results are robust to varying parameters governing the cost of emission reduction, economic damages associated with emissions, and households' liquidity valuation. In a second set of robustness checks, we modify the sectoral structure of the production side. Third, we consider various model extensions concerning the financial sector, such as a lower intermediation efficiency, imperfect diversification of banks across sectors, and the presence of a risk-free reserve asset. Lastly, we add the possibility of bond financing for fossil to the model and show that the efficacy of tilted bank capital regulation is even smaller in this case.

Lastly, we quantify the macroeconomic and climate impact of different regulatory policies at the onset of the net zero transition. Specifically, assume that global emission taxes increase linearly from their 2025 level of 5\$/tCO2 to 100\$/tCO2 in 2050, which reduces emissions to zero in our model and is consistent with the Paris agreement. In the short run, output drops substantially, while inflation and output gap spike, consistent with the idea of a negative supply shock. To assess whether tilted bank capital regulation can mitigate these adverse consequences, we also consider the case where the sustainability linked capital requirement is implemented alongside emission taxes. This slightly reduces marginal costs in the firm sector and hence softens the inflationary impact of the transition. The effect is however small in comparison to the initial inflationary surge, suggesting that tilted bank capital regulation can do little to limit the short run losses of ambitious climate policy action.

Related Literature. Our paper is related to two fast growing strands of literature. First, we contribute to the literature on interactions between financial frictions, climate change, and climate policy. There are several theoretical results on the relevance of financial frictions for the conduct of environmental or climate policy. Heider and Inderst (2022) and Döttling and Rola-Janicka (2025) show that financial frictions impair the conduct of climate policy and that the optimal Pigouvian emission tax is lower than in the absence of financial frictions, resonating with results on second-best regulatory policies by Davila and Walther (2022). Oehmke and Opp (2025) argue that tilted bank capital regulation are an ill-suited instrument to initiate a transition to net zero, consistent with our findings. The limited real effects of such policies is also in line with recent empirical work by Miguel et al. (2024).

We also contribute to the literature on financial and climate policies in quantitative E-DSGE models that study green QE (Ferrari and Nispi Landi, 2023) and green collateral policy (Giovanardi et al., 2023). Similar to our results, these papers point to a limited effectiveness of tilted central bank policies as climate policy instrument. Fornaro et al. (2024) propose a two-sector model with endogenous technological change and demonstrate that investment in the clean sector is more sensitive to interest rates. While firms in our setup would respond symmetrically to interest rate changes, we focus on financing frictions in the production sector to study the effects of tilted bank capital regulation. Taking a macroprudential angle, Carattini et al. (2023) study the effects of socially inefficient asset stranding on the macroeconomy through the lenses of a financial accelerator model. Annicchiarico et al. (2023) propose a model with climate policy where external financing constraints at the firm level amplify business-cycle fluctuations, such that (green) macroprudential policy operates as stabilization instrument. More generally, studying aggregate implications of sector-specific shocks goes back to at least Horvath (2000). To the best of our knowledge, there is no analysis of optimal bank capital regulation in multi-sector DSGE models. The optimal policy results presented in this paper, therefore, are not restricted to the effects of climate policy along the net zero transition, but also apply to sectoral re-allocations more generally.

Outline. Section 2 sets up the model and we describe its calibration in Section 3. In Section 4, we present the quantitative effects of tilted capital requirements, while discuss the economic intuition behind these results in Section 5. Optimal capital requirements in response to emission taxes and climate policy uncertainty are presented in Section 6. Section 7 demonstrates the robustness of our results. Lastly, Section 8 shows how tilted capital requirements affect the transition to net zero and discusses implications for short run policy trade-offs. Section 9 concludes.

2 Model

Time is discrete and indexed by $t = 1, 2, ...$. The model features households, a continuum of competitive banks, monopolistically competitive final good producers, three sectors of competitive intermediate good firms, indexed by $s \in \{c, f, n\}$, and sector-specific investment good producers. The intermediate firms produce non-energy (n), fossil energy (f), and clean energy (c) goods and finance their capital with equity and bank loans. The final good producer combines all intermediate goods with labor and is subject to nominal rigidities. Banks raise nominal deposits from households and extend nominal loans to intermediate good producers. The model is closed by a public sector levying emission taxes, conducting monetary policy and setting bank capital requirements.

Households. Each household i supplies labor in monopolistically competitive fashion at the real wage w_t and, following Smets and Wouters (2003), a share Ψ_N of households can not re-optimize their wage. Aggregate labor and consumption are denoted by n_t and c_t, respectively. Deposits held from time $t-1$ to t earn the nominal interest rate r^D_{t-1}, such that the budget constraint reads $c_t + d_{t+1} = w_t n_t + (1 + r^D_{t-1})\frac{d_t}{\pi_t} + div_t + T_t$, where div_t collects real dividends from banks and firms and T_t is a lump sum transfer from the government. We denote households' time discount factor by β. Households derive utility from consumption c_t and from holding end-of-period deposits d_{t+1}, and disutility from labor supply:

$$u(c_t, n_t, d_{t+1}) = \frac{c_t^{1-\gamma_C}}{1-\gamma_C} - \omega_N \frac{n_t^{1+\gamma_N}}{1+\gamma_N} + \omega_D \frac{d_{t+1}^{1-\gamma_D}}{1-\gamma_D} .$$

In Appendix A.1, we derive households' labor supply decision and a measure of wage dispersion that affects welfare. Solving the maximization problem also yields the Euler equation for deposits

$$1 = \mathbb{E}_t \Lambda_{t,t+1} \frac{1 + r^D_t}{\pi_{t+1}} + \omega_D \frac{d_{t+1}^{-\gamma_D}}{c_t^{-\gamma_C}} . \tag{1}$$

Here, $\Lambda_{t,t+1} \equiv \beta \frac{c_{t+1}^{-\gamma_C}}{c_t^{-\gamma_C}}$ is households' stochastic discount factor (sdf). The nominal risk-free rate r_t is implicitly defined by the sdf:

$$1 = \mathbb{E}_t \Lambda_{t,t+1} \frac{1 + r_t}{\pi_{t+1}} . \tag{2}$$

As deposits provide utility services to households, the deposit rate will always be below the risk-free rate and we refer to the (negative) spread $r^D_t - r_t$ as the liquidity premium. In our setup, bank deposits are nominally safe assets since depositors are protected from the risk of bank failure by public bailout guarantees, described in the next paragraph.[3]

Banks. Banks enter period t with liabilities $(1 + r^D_{t-1})\frac{d_t}{\pi_t}$ and sector-specific loans l^s_t granted last period. The bank-specific realized real payoff from its loan portfolio is given by $\mu_t \sum_s \mathcal{R}^s_t l^s_t$ and contains two components. First, it depends on sector-specific real loan payoffs that are denoted by $\mathcal{R}^s_t$. These payoffs in turn depend on (potentially) sector-specific default rates in the non-financial sector (described below). Second, banks

[3] Money in the utility specifications have been used in macroeconomics since the contribution by Poterba and Rotemberg (1986), see Drechsler et al. (2017), Begenau (2020) and Jamilov and Monacelli (2025) for recent examples.

are subject to uninsurable idiosyncratic bank risk shocks μ_t, which follow an i.i.d. log-normal distribution with standard deviation ς_μ and mean $-\varsigma_\mu^2/2$ to ensure that they have an expected value of one. These shocks reflect unmodeled limits to diversification of loan portfolios, consistent with empirical findings by Galaasen et al. (2021) and are a common modeling approach in the macro banking literature, see Mendicino et al. (2024) and the references therein.

If the idiosyncratic bank risk shock realizes below a threshold level $\overline{\mu}_t$, an individual bank is unable to service depositors and fails. It transfers all assets and liabilities to the fiscal authority who covers the shortfall and pays back depositors in full, while bankers are protected by limited liability. We follow Mendicino et al. (2024) in assuming that the fiscal authority incurs a resource losses from bailing out banks that is proportional to the amount of deposits manged by the fiscal authority, $T_t^{bail} = \zeta \cdot F(\overline{\mu}_t) \cdot \frac{d_t}{\pi_t}$, where $F(\overline{\mu}_t) \equiv \int_0^{\overline{\mu}_t} dF(\mu_t)$ is the share of failing banks in period t.

The default threshold is given by the risk shock realization $\overline{\mu}_t$ that enables the bank to exactly repay depositors:

$$\overline{\mu}_t = \frac{(1 + r_{t-1}^D)\frac{d_t}{\pi_t}}{\sum_s \mathcal{R}_t^s l_t^s} \ . \tag{3}$$

The within-period timing assumption is that loan payoffs realize and deposits from the previous period have to be repaid before the markets for new deposits d_{t+1} and new loans l_{t+1}^s open, such that banks can not raise new equity to repay liabilities from legacy deposits. However, we assume that banks are restructured intermediately after a failure so that they can extend new loans to firms at price q_t^s and raise equity and deposits from households at the end of period t. This facilitates aggregation into a representative bank.

Each bank maximizes shareholder value $\mathbb{E}_t \sum_{\tau=0}^{\infty} \Lambda_{t,t+\tau} \cdot div_{t+\tau}^b$, where $\Lambda_{t,t+\tau}$ is the sdf of bank owners (households) that each bank takes as exogenously given. The dividend of each bank in period t depends on its idiosyncratic shock realization μ_t and is given by

$$div_t^b = \mathbb{1}\{\mu_t > \overline{\mu}_t\}\Big(\mu_t \sum_s \mathcal{R}_t^s l_t^s - (1 + r_{t-1}^D)\frac{d_t}{\pi_t}\Big) + d_{t+1} - \sum_s q_t^s l_{t+1}^s \ ,$$

where the indicator $\mathbb{1}\{\mu_t > \overline{\mu}_t\}$ captures that a bank only receives the loan payoff net of deposit repayments if it does not fail. While banks are heterogeneous *ex post*, all banks are *ex ante* identical at the end of each period when they make their loan pricing decision, due to the immediate restructuring assumption and the i.i.d. assumption on the bank risk shock. We discuss potential implications of *ex ante* heterogeneity in Section 7.

Banks are required to finance a (potentially sector-specific) fraction κ_t^s of their loans with equity:

$$\mathbb{E}_t \frac{(1+r_t^D)d_{t+1}}{\pi_{t+1}} \leq \sum_s (1-\kappa_t^s)\mathbb{E}_t \mathcal{R}_{t+1}^s l_{t+1}^s \ . \tag{4}$$

We show in Appendix A.2 that this constraint is binding in all states, see also Begenau (2020). Intuitively, deposit financing is cheaper than equity due to the bailout guarantee and liquidity premia on deposits. Furthermore, each atomistic bank has no effect on the sdf $\Lambda_{t,r+1}$ of its owners such that we can abstract from precautionary motives and banks behave as if they were risk neutral.

Hence, we solve the profit maximization problem subject to the binding capital constraint (4) and obtain the loan pricing condition

$$q_t^s = \mathbb{E}_t \Bigg((1-\kappa_t^s) \underbrace{\left(\frac{\pi_{t+1}}{1+r_t^D} - \Lambda_{t,t+1}\big(1 - F(\overline{\mu}_{t+1})\big) \right)}_{\text{Deposit Financing Wedge } \Xi_{t+1}} + \underbrace{\Lambda_{t,t+1}\Big(1 - G(\overline{\mu}_{t+1})\Big)}_{\text{Banker sdf } \overline{\Lambda}_{t,t+1}} \Bigg) \mathcal{R}_{t+1}^s \ . \tag{5}$$

Its detailed derivation is relegated to Appendix A.2. The loan pricing condition connects the expected real loan payoff $\mathcal{R}_{t+1}^s$ to the loan price via the bank sdf, which is given by the term in parentheses. The bank sdf in turn consists of the benefits of deposit financing Ξ_{t+1}, weighted by the deposit financed loan share $(1-\kappa_t^s)$, and the banker sdf $\overline{\Lambda}_{t,t+1}$. The loan pricing condition takes the possibility of bank failure into account, as banks only receive the loan payoff if they do not fail. This is reflected by the expected bank productivity conditional on not failing $1 - G(\overline{\mu}_{t+1}) \equiv \int_{\overline{\mu}_{t+1}}^{\infty} \mu_{t+1} dF(\mu_{t+1})$.[4]

[4] If banks were fully equity financed ($\kappa_t^s = 1$ for all s), the deposit financing wedge Ξ_t would be irrelevant for the loan pricing condition. Furthermore, the bank failure rate would be zero, i.e. the banker sdf coincides with the household sdf. Consequently, the loan pricing condition would collapse to a standard consumption based asset pricing condition $q_t^s = \mathbb{E}_t\left[\Lambda_{t,t+1}\mathcal{R}_{t+1}^s\right]$, rather than an intermediary asset pricing condition. In contrast, if there were no capital requirements ($\kappa_t^s = 0$ for all s), bank

The deposit financing wedge shows that both the bailout put and liquidity services affect the pricing of loans via bankers' sdf. The deposit financing wedge Ξ_t reflects the benefit of financing a loan through deposits $r_t^D < r_t$ due to their liquidity benefits. It also reflects the bailout put, i.e. the expected repayment obligation in period $t+1$ of raising one unit of deposits in period t is only $1 - F(\overline{\mu}_{t+1})$. All else equal, loan prices increase if capital requirements are relaxed, since this allows banks to increase the deposit financed share.

The banker sdf $\overline{\Lambda}_{t,t+1}$ is closely related to the household sdf due to the perfect risk-sharing assumption, but also contains the expected bank profitability conditional on not failing, which is given by $1 - G(\overline{\mu}_{t+1})$ and is smaller than one. Intuitively, the bank loses control over its assets if it fails, such that expected loan payoff is discounted more strongly if bank failure is more likely. This mechanism has been referred to as "guarantee overhang" (Bahaj and Malherbe, 2020) and stringent capital requirements also have a positive effect on loan prices by increasing the term $(1 - G(\overline{\mu}_{t+1}))$.

Intermediate Good Firms. We now describe the demand side for corporate loans that links bank capital regulation to the real sector and to climate policy. There are three types of intermediate good firms which produce a homogeneous sector-specific intermediate good using a linear production technology $z_t^s = m_t k_t^s$. Here, m_t is an uninsurable idiosyncratic productivity shock that realizes at the beginning of each period (before any decisions are made) and is i.i.d. log-normally distributed with standard deviation ς_M. We normalize its mean parameter to $-\varsigma_M^2/2$, which ensures that the shock has an expected value of one. Capital can be taxed at the constant rate τ^s and has to be acquired from investment good firms that are subject to investment adjustment costs at the sector-specific price of capital ψ_t^s. The supply of investment goods i_t^s as a function of its price is derived in Appendix A.4. We only present the maximization problem of the fossil energy firm, as the problems for clean energy and non-energy firms are identical, with the exception that those firms do not have to pay emission taxes.

failure rate and deposit supply would be very high. Since we assume that fiscal losses during a bailout are proportional to bank deposits and the household utility function is concave in deposits, there is an interior solution for the optimal (symmetric) long run capital requirement.

Firms are managed on behalf of households by impatient managers with time preference parameter $\widetilde{\beta} < \beta$. Firm managers maximize the present value of dividends, discounted by a sdf $\widetilde{\Lambda}_{t,t+1} \equiv \widetilde{\beta}\frac{c_{t+1}^{-\gamma_C}}{c_t^{-\gamma_C}}$ which reflects the relative impatience of firm managers and gives rise to a borrowing motive. To finance their investment, firms can either use equity (which reduces current dividends) or nominal long-term loans l_t^s (which reduces expected future dividends). A share $0 < \chi \leq 1$ of loans matures each period and the non-maturing share $(1-\chi)$ is rolled over at the current loan price.

Idiosyncratic productivity shocks imply that borrowing is risky. Firms default on the maturing share χl_t^f, if revenues from production $\widetilde{p}_t^f m_t z_t^f$ fall below the required loan repayment χl_t^f, where $\widetilde{p}_t^f$ is the fossil energy price net of any expenses related to emission taxes, described below. We define the threshold realization of the idiosyncratic productivity shock below which a firm defaults as:

$$\overline{m}_{t+1}^f \equiv \frac{\chi l_{t+1}^f}{\pi_{t+1}\widetilde{p}_{t+1}^f k_{t+1}^f} . \tag{6}$$

In the default case, banks are entitled to the period t output, but have to pay a restructuring costs φl_t^f. Following Gomes et al. (2016), we assume that firms are restructured immediately which, together with the i.i.d. nature of idiosyncratic productivity shocks, permits aggregation into a representative firm in each sector.

The expected productivity of a defaulting firm is defined as $G(\overline{m}_t^f) \equiv \int_0^{\overline{m}_t^f} m dF(m)$ while the default probability is given by $F(\overline{m}_t^f) \equiv \int_0^{\overline{m}_t^f} dF(m)$. We integrate out the idiosyncratic shock to get the average per-unit loan payoff in the fossil sector:

$$\mathcal{R}_t^f = (1-\chi)q_t^f + \frac{\chi}{\pi_t}\left(1 - F(\overline{m}_t^f) + \frac{G(\overline{m}_t^f)}{\overline{m}_t^f} - F(\overline{m}_t^f)\varphi\right) . \tag{7}$$

The first term is the share $(1-\chi)$ of loans that are rolled over, valued at market price q_t^f. The second term represents the payoff from maturing share χ. It consists of the repayment probability $1 - F(\overline{m}_t^f)$, the production revenues seized in case of default $\frac{G(\overline{m}_t^f)}{\overline{m}_t^f}$, and expected restructuring cost $F(\overline{m}_t^f)\varphi$.

Emission Abatement. Fossil energy firms are subject to emission taxes τ_t^e. We follow Heutel (2012) in assuming that unabated emissions are proportional to production and allow fossil energy firms to undertake a costly abatement effort a_t. Emissions are therefore given by $e_t = (1 - a_t)z_t^f$ while the emission tax payed in period t is given by $\tau_t^e(1 - a_t)z_t^f$. Alternatively, the variable a_t can be interpreted as the share of fossil firms switching to a less productive clean energy technology (Acemoglu et al., 2012). Abatement costs are convex in the abatement effort and proportional to output $\frac{\theta_1}{\theta_2+1}a_t^{\theta_2+1}z_t^f$ with θ_1, $\theta_2 > 0$. Hence we can write the fossil energy price net of per-unit emission taxes and abatement cost as

$$\widetilde{p}_t^f = p_t^f - \tau_t^e(1 - a_t) - \frac{\theta_1}{\theta_2 + 1}a_t^{\theta_2+1} \; . \tag{8}$$

All else equal, higher emission tax raise the break-even productivity shock realization $\overline{m}_t^f$ below which the firm defaults. Banks' loan pricing condition takes this into account ex ante, such that fossil energy firms adjust their investment and leverage choices.

We also consider sustainability linked capital requirements (SLC) as an alternative policy to support the net zero transition. Here, we assume that the fossil capital requirement depends linearly on the abatement effort a_t, such that fossil energy firms are rewarded with a higher loan price for undertaking a greater abatement effort:

$$\kappa_t^f = (1 - a_t) \cdot \kappa_t^{pen} + a_t \cdot \kappa_t^{reward} \; . \tag{9}$$

The most stringent SLC is given by κ_t^{pen}=1 so that fossil loans receive a capital requirement of 100% if firms do not abate (a_t=0). Firms receive a capital requirement $\kappa_t^{reward} < \kappa_t^{pen}$ under full abatement.

Maximization Problem. Firm managers choose abatement effort, investment and loans to maximize shareholder value,

$$\max_{\{a^f_{t+s}, i^f_{t+s}, l^f_{t+s}\}} \quad \mathbb{E}_t \sum_{s=0}^{\infty} \widetilde{\Lambda}_{t,t+s} div^f_{t+s}\ , \quad \text{where} \tag{10}$$

$$\begin{aligned} div^f_{t+s} = 1\{m_{t+s} > \overline{m}^f_{t+s}\}\Big(\widetilde{p}^f_{t+s} m_{t+s} k^f_{t+s} - \chi \frac{l^f_{t+s}}{\pi_{t+s}}\Big) \\ - (1+\tau^f)\psi^f_{t+s} i^f_{t+s} + q(\overline{m}^f_{t+s+1}, a_{t+s})\Big(l^f_{t+s+1} - (1-\chi)\frac{l^f_{t+s}}{\pi_{t+s}}\Big)\ , \end{aligned}$$

subject to the law of motion for capital $i^f_t = k^f_{t+1} - (1-\delta_K)k^f_t$, banks' loan pricing condition (5), the default threshold (6), the loan payoff (7), the relationship between the abatement effort and net fossil energy price (8), and to the fossil capital requirement (9) if the SLC is active ($\kappa^{pen}_t \neq \kappa^{reward}_t$). Because abatement costs and emission taxes are proportional to current output and firms can adjust a_t each period, *every* firm chooses the same a^*_t - including defaulters that are managed by the bank in the default period. Optimal abatement is given by

$$a^*_t = \left(\frac{\tau^e_t + \frac{\partial q^f_t}{\partial a_t}\big(l^f_{t+1} - (1-\chi)\frac{l^f_t}{\pi_t}\big)\frac{1}{z^f_t}}{\theta_1} \right)^{1/\theta_2}\ , \tag{11}$$

which increases in the emission tax and potentially on the SLC through the partial derivative of the loan price schedule $\frac{\partial q^f_t}{\partial a_t} = \mathbb{E}_t\big(\kappa^{pen}_t - \kappa^{reward}_t\big)\Xi_{t+1}\mathcal{R}^f_{t+1}$. The optimal abatement effort collapses to the familiar expression $a^*_t = \left(\frac{\tau^e_t}{\theta_1}\right)^{\frac{1}{\theta_2}}$ if $\kappa^{pen}_t = \kappa^{reward}_t$.

Optimal loans and investment jointly solve a dynamic maximization problem and firms take the effect of their inter-temporal choices on future loan prices into account, which we make explicit in the following. Denoting the Lagrange-multiplier on the default threshold by λ^f_t, the first-order conditions for loans and investment are given by

$$q(\overline{m}^f_{t+1}, a_t) - \lambda^f_t \frac{\overline{m}^f_{t+1}}{l^f_{t+1}} = \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}}\Big(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q(\overline{m}^f_{t+2}, a_{t+1})\Big)\ , \tag{12}$$

and

$$\psi^f_t(1+\tau^f) - \lambda^f_t \frac{\overline{m}^f_{t+1}}{k^f_{t+1}} = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\psi^f_{t+1}(1-\delta_K)(1+\tau^f) + \widetilde{p}^f_{t+1}\Big(1 - G(\overline{m}^f_{t+1})\Big)\Big)\ , \tag{13}$$

whereas the risk choice $\overline{m}^f_{t+1}$ is determined according to

$$-\lambda^f_t - \frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \left(l^f_{t+1} - (1-\chi) \frac{l^f_t}{\pi_t} \right) = \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} \left(\left(l^f_{t+2} - (1-\chi) \frac{l^f_{t+1}}{\pi_{t+1}} \right) \frac{\partial q^f_{t+1}}{\partial \overline{m}^f_{t+2}} \frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}} \right) . \tag{14}$$

Additional details to the maximization problem and its solution are provided in Appendix A.3. The first-order condition for loans (12) equates the marginal benefit of taking up a loan with its costs. Since the loan maturity exceeds one period, it also contains a dilution term that enters through the multiplier on the default threshold λ^f_t, which is in turn pinned down by equation (14). The cost of marginally increasing loans consist of the share χ of loans that matures next period, weighted by the repayment probability, and the roll-over part $(1-\chi)$, valued by next period's loan price $q(\overline{m}^f_{t+2}, a_{t+1})$.

Equation (13) requires that the cost of purchasing one unit of capital $(1+\tau^f)\psi^f_t$ equals the expected discounted payoff. The payoff consists of the value of undepreciated capital next period $\psi^f_{t+1}(1-\delta_K)(1+\tau^f)$ and expected revenues per unit of capital $\widetilde{p}^f_{t+1}$, i.e. the fossil energy price net of emission taxes and abatement costs, and multiplied by the expected productivity conditional on repayment $\left(1 - G(\overline{m}^f_{t+1})\right)$. The term $\lambda^f_t \frac{\overline{m}^f_{t+1}}{k^f_{t+1}}$ on the LHS captures that investment reduces the default probability in period $t+1$ and hence raises the loan price in period t, which increases current dividends. Capital has a dual purpose in this model, as it also creates debt capacity for intermediate good firms.

Final Good Firms. Monopolistically competitive firms aggregate both energy inputs and the non-energy intermediate good together with labor into the final good y_t according to a nested CES-structure:

$$y_t = A_t \exp\{-\psi_E E_t\} \widetilde{z}^{\alpha}_t n^{1-\alpha}_t \ , \tag{15}$$

where exogenous total factor productivity A_t follows $\log(A_t) = \rho_A \log(A_{t-1}) + \sigma_A \epsilon^A_t$ with $\epsilon^A_t \sim N(0,1)$ and α governs the labor share. The term $\exp\{-\psi_E E_t\}$ reflects productivity losses from climate change damages, where the parameter ψ_E governs the damage-to-GDP ratio. This is a slightly simplified version of the macro-climate model of Golosov et al. (2014), which abstracts from explicitly modelling global temperature dynamics. The intermediate goods bundle is defined as

$$\widetilde{z}_t = \left(\widetilde{\nu}(z_t^e)^{\frac{\widetilde{\epsilon}-1}{\widetilde{\epsilon}}} + (1-\widetilde{\nu})(z_t^n)^{\frac{\widetilde{\epsilon}-1}{\widetilde{\epsilon}}}\right)^{\frac{\widetilde{\epsilon}}{\widetilde{\epsilon}-1}} , \tag{16}$$

where $\widetilde{\nu}$ is the weight on energy and $\widetilde{\epsilon}$ is the substitution elasticity between energy and non-energy goods. The energy bundle is given by

$$z_t^e = \left(\nu(z_t^c)^{\frac{\epsilon-1}{\epsilon}} + (1-\nu)(z_t^f)^{\frac{\epsilon-1}{\epsilon}}\right)^{\frac{\epsilon}{\epsilon-1}} , \tag{17}$$

where ν is the clean energy weight and ϵ is the substitution elasticity between fossil and clean energy. Solving the profit maximization problem yields standard demand functions for labor and intermediate goods. Each firm i sells its final good variety to households at a markup over its marginal costs, subject to price adjustment costs (Rotemberg, 1982). This gives rise to a New Keynesian Phillips curve linking monetary policy to the production sector. We relegate the analytical steps to Appendix A.5. The model is closed by assuming that emission tax revenues and losses from bank bailouts are rebated in lump-sum fashion to households ($T_t = \tau_t^e e_t - T_t^{bail}$) and that the central bank sets the nominal interest rate according to a Taylor-type rule:

$$1 + r_t = (1 + r^{SS})\pi_t^{\varphi_\pi} + \sigma_R \epsilon_t^R \ , \quad \text{where } \epsilon_t^R \sim N(0,1) \tag{18}$$

The resource constraint is given by

$$y_t = c_t + \sum_s i_t^s \Big(1 + \frac{\Psi_I}{2}(\frac{i_t^s}{i_{t-1}^s} - 1)^2\Big) + \frac{\Psi_P}{2}(\pi_t - 1)^2 + \frac{\theta_1}{\theta_2 + 1}\left(\frac{\tau_t^e}{\theta_1}\right)^{\frac{\theta_2+1}{\theta_2}} z_t^f + \sum_s \chi\varphi F(\overline{m}_t^s) l_t^s + \zeta F(\overline{\mu}_t) d_t \ ,$$

and reflects the cost of corporate default and bank failure, as well as emission abatement and productivity losses from emissions through the aggregate production function (15). Liquidity benefits of deposits do not enter the resource constraint but are part of the welfare objective via the household utility function.

3 Calibration

The model is calibrated to financial market data from the US and each period corresponds to one year. All data moments are based on the sample from 1984 to 2013. While historical bank failure rates are publicly available only after 1984, the sample truncation point in 2013 implies that our data sample does not cover the period after the Paris agreement, which was signed in 2015. This justifies a symmetric calibration of intermediate good producers, setting the emission tax to zero, and imposing an identical capital requirement across sectors ($\kappa_t^c = \kappa_t^f = \kappa_t^n = \kappa^{sym}$).

Households. We fix household's consumption CRRA parameter to γ_C=2, the labor disutility curvature γ_N=1, and the weight on labor disutility to ω_N=8 to imply a steady state labor supply n^{SS}=0.3. The household discount factor is set to β=0.99, implying an annualized real interest rate of 1%. We set $\widetilde{\phi}$=3, which implies a steady state wage markup of 50% (Smets and Wouters, 2003). The share of sticky wage households is set to Ψ_N=0.63, following Del Negro et al. (2015).

The weighting of household's valuation of liquidity services ω_D=0.007 is calibrated to match the liquidity premium, defined as the average deposit spread below the risk-free rate $r^D - r$. We target a liquidity premium of -40bps (van Binsbergen et al., 2022; Krishnamurthy and Li, 2022). The curvature is set to γ_D=1.5, targeting the excess volatility of liquid asset holdings $\sigma(d/y)$ over the volatility of GDP $\sigma(y)$, which is given by 0.98 in the data (Lucas and Nicolini, 2015). A very high curvature would imply that households hold an approximately constant amount of deposits relative to their income and hence reduce the excess volatility of deposit holdings $\sigma(d/y)/\sigma(y)$ to zero.

Banks. The next set of parameters concerns banks. Historical bank failure rates are computed based on data from the Federal Deposit Insurance Corporation (FDIC). Following Elenev et al. (2021), we compute the bank failure rate as the share of assisted mergers and banks in resolution. This yields an average bank failure rate of 0.7% per year, which obtains in our model by setting the standard deviation of banks' risk shock to ς_μ=0.0325. We set the bailout loss parameter to ζ=0.011, which renders a symmetric long run capital requirement of 8% optimal from a utilitarian welfare perspective.[5]

Intermediate Good Firms. The parameters governing financial frictions of intermediate good firms are set to match moments typically used in the macro-finance literature. We set the annual capital depreciation rate to a standard value of δ_K=0.08 and the average loan maturity to five years, implying χ=0.2. We jointly calibrate the standard deviation of idiosyncratic productivity shocks ς_M=0.2, the firm owner discount factor $\widetilde{\beta}$=0.988 and the restructuring cost parameter φ=0.12 to match the corporate default rate $F(\overline{m})$, the leverage ratio ($lev \equiv l/k$), and the recovery rate on loans, which we formally define in Appendix A.3. Our parameterization implies recovery rate of 82%, which corresponds to the average debt recovery rate for the US over time, based on the dataset provided by Djankov et al. (2008). We target a corporate default rate of 1.7% and a leverage ratio of 30% at book values. The data moments correspond to the average default rate of rated firms and to the ratio of corporate sector net debt over net assets in the US from 1984-2013, respectively.

Final Producers. The Cobb-Douglas coefficient in the production function of final good producers is fixed at α=1/3. The elasticity of substitution between final goods varieties ϕ=6 implies a markup of 20%. We set the Rotemberg price adjustment parameter to Ψ_P=4.5, which implies an average duration of five quarters. The central bank reaction to inflation ϕ_π=2 and the standard deviation of monetary policy shocks σ_R=0.02 jointly match the excess volatility of inflation and short-term interest rates over GDP. The investment adjustment cost parameter is set to Ψ_I=2, targeting the excess volatility of investment over GDP. Lastly, persistence and standard deviation of the aggregate TFP shock are set to ρ_A=0.8 and σ_A=0.004, which are standard values in the business cycle literature. Since we only target excess volatilities of endogenous objects over the volatil-

[5] Specifically, we solve the model in its second-order approximation around the deterministic steady state over a discrete grid of capital requirements κ^{sym} and evaluate household welfare for each κ^{sym}. The welfare function is based on households' period utility and takes into account monopolistically competitive labor supply, see equation (A.31) in Appendix A.1 for further details. While we do not consider Ramsey optimal policy throughout the paper, it is worth noting that the long run Ramsey-optimal capital requirement is quantitatively very similar.

ity of GDP, the parameters of the TFP shock process are not central to our calibration strategy.

Sectoral Shares. Within the energy sector, we set ν=0.3865 implying that 20% of firms in the energy sector use the clean technology initially, which is a standard value used in comparable models (Giovanardi et al., 2023; Ferrari and Nispi Landi, 2023; Barnett, 2024) and consistent with recent data from the EU and the US. We fix the substitution elasticity between clean and fossil energy in equation (17) at ϵ=3, based on empirical work by Papageorgiou et al. (2017). This value is commonly used in multi-sector DSGE models, see for example Fried et al. (2022) or Coenen et al. (2024).

The weighting parameter on the energy bundle in the intermediate good (16) is set to $\widetilde{\nu}$=0.0015, which implies an energy share of 10%. This value is for example used by Bartocci et al. (2024) in a calibration to the EU and is also consistent with an 8% ratio of fossil energy to total capital (Campiglio et al., 2024a). The elasticity $\widetilde{\epsilon}$=0.2 between energy and non-energy goods in equation (16) strikes a middle ground between very small short run elasticities and the directed technological change literature, which typically assumes that technology is Cobb-Douglas in the longer run, see Hassler et al. (2021) for a discussion. Our parameter choice coincides with Bartocci et al. (2024) who calibrate a large scale DSGE model to sectoral data from the EU. Choosing these comparatively low sectoral substitution elasticities and high weights on fossil energy appears to be a reasonable upper bound for the adverse side effects of climate policy on bank liquidity provision. In Section 7, we show that all our quantitative results are robust to using higher elasticities and lower fossil energy weights.

Climate Block. The curvature parameter of the abatement cost function is set to θ_2=1.6, which is a standard value in the literature (Heutel, 2012). The scale parameter is set to θ_1=0.05 to ensure that an emission tax of 100$/tCO2 implements net zero emissions. We refer to Appendix C for details on how to convert model taxes into $/tCO2. This value is consistent with similar E-DSGE models (see e.g. Ferrari and Nispi Landi, 2023) and implies that 1.15% of GDP are spent on abatement under net zero emissions. We

Moment	**Model**	**Data**
Targeted Means		
Liquidity premium $ave(s^D)$ in bps	-40	-40
Corporate leverage $ave(lev)$ in %	29	26
Corporate default $ave(F(\overline{m}))$ in %	1.7	1.7
Recovery rate $ave(recov)$ in %	81	82
Bank failure rate $ave(F(\overline{\mu}))$ (%)	0.59	0.6
Targeted Excess Volatilities		
Relative vol. investment $\sigma(i)/\sigma(y)$	1.57	1.50
Relative vol. deposit/GDP $\sigma(d/y)/\sigma(y)$	1.05	0.97
Relative vol. interest rate $\sigma(r)/\sigma(y)$	1.41	1.02
Relative vol. inflation $\sigma(\pi)/\sigma(y)$	0.41	0.68
Untargeted Excess Volatilities		
Relative vol. consumption $\sigma(c)/\sigma(y)$	0.92	0.88
Relative vol. leverage $\sigma(lev)/\sigma(y)$	0.49	0.93
Relative vol. corporate default $\sigma(F(\overline{m}))/\sigma(y)$	0.40	0.86
Relative vol. bank failure $\sigma(F(\overline{\mu}))/\sigma(y)$	0.46	0.33
Untargeted Correlations		
Emissions-GDP $cor(y, e)$	0.41	0.78
Deposit/GDP-GDP $cor(y, d/y)$	-0.22	-0.22
Markdown-GDP $cor(y, \mu)$	-0.35	-0.47
Leverage-GDP $cor(y, lev)$	-0.63	-0.58
Firm default-GDP $cor(y, F(\overline{m}))$	-0.76	-0.32
Bank failure-GDP $cor(y, F(\overline{\mu}))$	-0.83	-0.32
Firm default-Inflation $cor(\pi, F(\overline{m}))$	-0.06	-0.33
Bank failure-Inflation $cor(\pi, F(\overline{\mu}))$	-0.02	-0.11
Firm default-Bank failure $cor(F(\overline{m}), F(\overline{\mu}))$	0.55	0.56

Table 1: Model Fit: All model-implied moments are computed based on a second-order approximation around the deterministic steady state. All moments are expressed in relative deviations from their steady state value. All data moments are based on real data and de-trended using an HP-filter with smoothing parameter of 6.25.

operationalize our analysis of climate policy uncertainty by using emission tax shocks:

$$\tau_t^e = (1 - \rho_T)\tau^e + \rho\tau_{t-1}^e + \sigma_\tau \epsilon_t^\tau \ , \tag{19}$$

where we set σ_τ=0.0025, which implies a standard deviation of 5\$/tCO2. This volatility is of a similar magnitude as the largest shocks to emission certificate prices in the European ETS (Berthold et al., 2023) and provides a reasonable upper bound. The parameter governing emission damages is set to Ψ_E=0.1, which implies a GDP loss of around 4%

in the absence of climate policies, which is in line with comparable E-DSGE models. We show that our results do not critically depend on climate parameters in Section 7. Our baseline parameterization is summarized in Table C.3 in Appendix C.

Model Fit. The upper two panels of Table 1 present the model fit to targeted first and second moments. The third panel demonstrates the model's ability to reconcile additional untargeted second moments typically used in the macro-finance and macro-banking literature. Consumption, leverage, corporate default and bank failure are all less volatile than GDP, consistent with the data. The fourth panel considers correlations of different variables with GDP and inflation. The pro-cyclicality of emissions is captured well by the model. Deposit holdings and markdowns are moderately counter-cyclical, consistent with the data. Furthermore, the model implies a counter-cyclical firm default probability. Similarly, the bank failure rate inherits the counter-cyclicality of corporate default. Consequently, bank failure and corporate default are positively correlated. Lastly, the bank failure and corporate default rates are negatively correlated with inflation due to the nominal denomination of deposits and loans.

4 Bank Regulation as Climate Policy Instrument

In this section, we use our calibrated model to study bank regulations as climate policy instrument. Figure 2 summarizes how tilted capital requirements (on the x-axis) affect key welfare-relevant variables in the long run. The dashed red line refers to the simple fossil penalizing capital requirement κ^f, while the dotted green line corresponds to the penalty capital requirement κ^{pen} in the SLC (9). Here, we fix κ_t^{reward}=8%, i.e. firms initially operating the fossil technology are treated like clean firms if they abate all their emissions ($a_t = 1$). The top left panel shows that simple fossil capital requirements only have a modest impact on emissions. At their maximum level of κ^f=100%, emissions are around 1% smaller than in the baseline without policies. By contrast, emissions decline by almost 5% for a stringent SLC with κ^{pen}=100%. This discrepancy arises because only the SLC induces abatement incentives to fossil firms, whereas emissions only decline by downsizing the fossil sector in the case of simple fossil capital requirements. Setting κ^f=100% resembles a "divestment" strategy induced by the regulator, which turns out to be ineffective at reducing emissions and comes with substantial adverse side effects, as we discuss next.

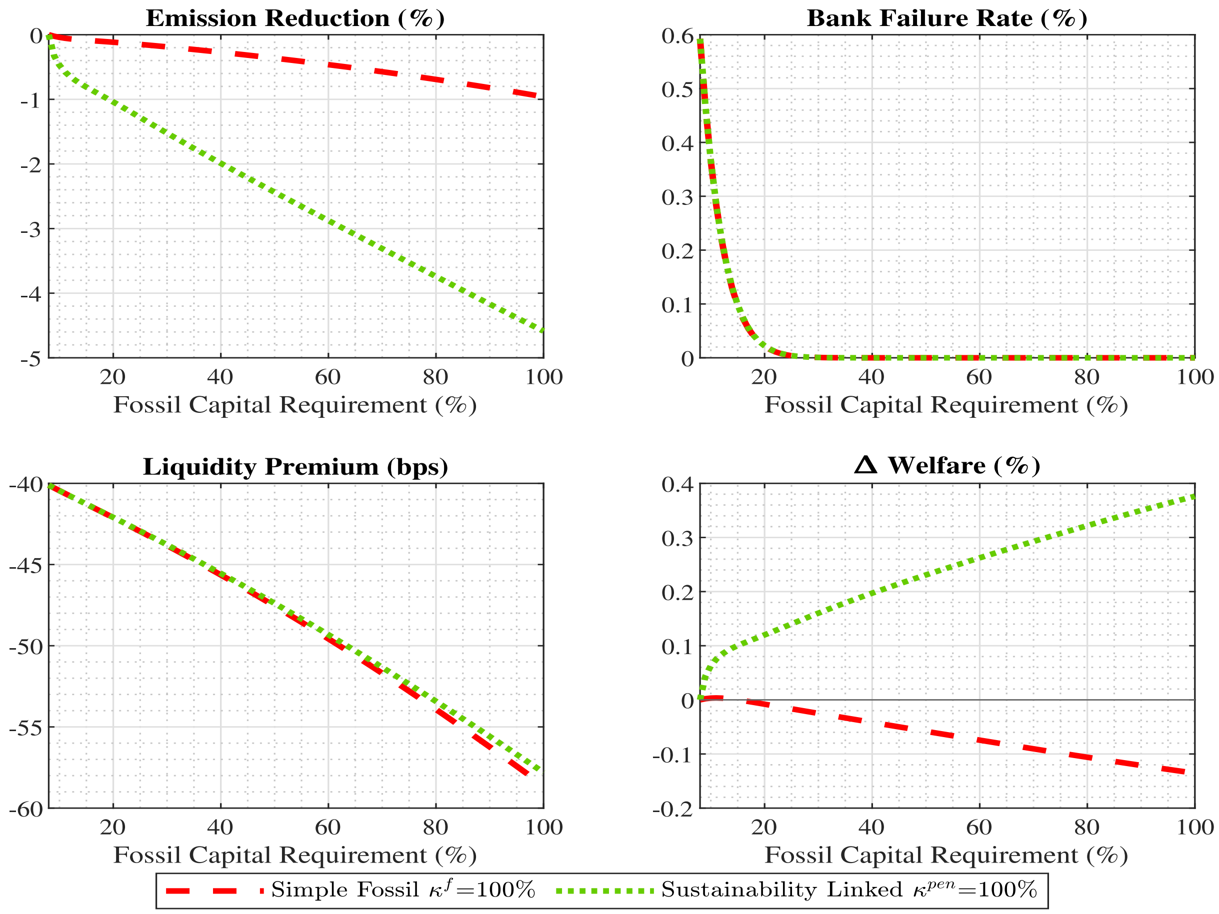

Figure 2: Long Run Effects of Fossil Capital Requirements. The upper left panel displays the share of abated emissions a_t for different κ_t^f (dashed red line) and different penalty requirements κ_t^{pen}, while fixing κ_t^{reward} = 8% (dotted green line). Welfare is defined in equation (A.31) and expressed in consumption equivalent relative to the baseline without climate policies.

In addition to reducing emissions, tilted bank capital requirements have two important implications for banks. First, requiring banks to hold more equity for fossil loans makes them safer, which is reflected in the much smaller bank failure rate even for modest increases in κ^f (top right panel of Figure 2). At the same time, this reduces the supply of deposits to households and widens the liquidity premium from -40bps to around -60bps (bottom left of Figure 2). The net welfare effect, which takes into account bank failure, liquidity supply and emission damages is shown in the bottom right panel. A lower bank failure rate has a positive welfare effect through the reduction of losses from restructuring banks, but this effect vanishes for higher capital requirements as banks are already safe. However, further increasing κ^f or κ^{pen} exacerbates the negative effects on liquidity supply. By the deposit channel of bank capital regulation, this results in a welfare loss. Tilted bank capital regulation faces a key trade-off between emission reduction and reducing liquidity supply. In Appendix D.1, we also discuss how the deposit contraction increases the loan pricing schedules and thereby provides risk-taking and investment incentives across *all* sectors.

Quantitatively, this trade-off is resolved heavily against the fossil penalizing capital requirement. Its net welfare effect is positive but small for κ^f between 8% and 15%. For higher values, the adverse effects on liquidity supply dominate and welfare declines if κ^f exceeds 15%. By contrast, sustainability linked capital requirements raise total welfare because they reduce emissions more effectively. Notably, deposits also contract by slightly less in this case, since fossil loans do not receive a 100% capital requirement but can at least be partially deposit financed. Such a policy-induced "engagement" strategy turns out to have a positive welfare effect that increases in κ^{pen}.

To put the effects of tilted bank capital regulation into perspective, it is helpful to benchmark tilted bank capital requirements against different fiscal instruments. The second column of Table 2 illustrates the implications of setting the fossil capital requirement to κ^f=100%. As we have already seen in Figure 2, the liquidity premium widens to -59bps and the bank failure rate declines to almost zero. Emissions decline by around one percent, which is comparable to the effects of a simple 1% tax on fossil capital, which we display in column (6) of Table 2. Such a fossil capital tax yields a comparable emission reduction and does not affect the abatement effort in the fossil sector. While it does not change the bank failure rates or the supply of deposits, it still reduces welfare by diminishing the incentives to accumulate capital.

The third column of Table 2 displays the effects of sustainability linked capital requirements. Without emission taxes in place, the optimal abatement effort turns out to be around 4.5%, such that banks have to finance almost 96% of fossil loans using equity that is costly to them. This still reduces the bank failure to almost zero and implies

Moment	(1) Baseline	(2) κ^f=100%	(3) SLC	(4) τ^e=5$	(5) τ^e=100$	(6) τ^f=1%
Δ Emissions	-	-0.96%	-4.59%	-17.15%	-100%	-1.05%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.59%
Liquidity Premium	-40bps	-59bps	-58bps	-41bps	-43bps	-40bps
Welfare Gain, CE	-	-0.14%	+0.38%	+1.99%	+6.95%	+0.00%

Table 2: Long Run Effects of Selected Policies. Column (1) reflects the baseline calibration, column (2) refers to a simple fossil penalizing capital requirement ($\kappa^f = 100\%$), column (3) to a sustainability linked capital requirement with $\kappa^{pen} = 100\%$ and $\kappa^{reward} = 8\%$, column (4) to a 5$/tCO2 emission tax, column (5) to the 100$/tCO2 emission tax and column (6) to a 1% tax on fossil capital. Welfare is defined in equation (A.31) and computed relative to the baseline without climate policies and expressed in consumption equivalents.

a drastic reduction in liquidity provision, which is reflected in the liquidity premium of -58bps. The welfare effect is positive at around 0.4% in consumption equivalents. This suggests that sustainability linked capital requirements are a more powerful climate policy instrument than simply increasing κ^f. Nevertheless, their effect on emissions amounts to merely a modest emission tax τ^e. In the fourth column, we show the effects of a 5$/tCO2 emission tax, which corresponds to the coverage-weighted global emission taxes in place in 2025, see Appendix C for details. Even this small tax delivers a much larger emission reduction of around 17% without materially affecting deposit supply and the bank failure rate. Consequently, the welfare gain is considerably larger at 1.99% in consumption equivalents. The fifth column provides the net zero emissions benchmark, which is achieved by setting the tax to 100$/tCO2.

5 The Transmission of Tilted Capital Requirements

To better understand how tilted bank capital regulation can affect emissions and why its effects are comparatively small, we consider a simplified version of the model.

Simplifying the Model. Throughout this section, we focus on the long run by setting A_t=1 for every t, and we drop the expectations operator and time index. To obtain analytically tractable first-order conditions, we consider the case of one period debt (χ=1) and assume that there are no debt restructuring cost (φ=0) but that all output is lost in the case of default. In Appendix B, we show that the simplified loan pricing schedule can be written as

$$q(\overline{m}^f_{t+1}) = \Big((1-\kappa^f_t)\Xi_{t+1} + \beta\Big)\Big(1 - F(\overline{m}^f_{t+1})\Big) \ , \tag{20}$$

Furthermore, there is full capital depreciation (δ_K=1) and no investment adjustment cost (Ψ_I=0) and we abstract from labor, non-energy goods and monopolistic competition in the aggregate production function. The first-order condition for fossil capital can then by approximated by

$$\psi^f - \frac{\partial q^f}{\partial k^f} l^f = \widetilde{\beta}\widetilde{p}^f \ , \tag{21}$$

This condition requires that the price of fossil capital equals its discounted expected payoff, i.e. the fossil energy price net of emission taxes and abatement costs $\widetilde{p}^f$. The additional term on the LHS of equation (21) captures the dual purpose of investment in our model. More capital increases the current loan price ($\frac{\partial q^f}{\partial k^f} > 0$) by making default less likely in the next period. This enables firms to front-load dividend payouts.

Simple Fossil Capital Requirements. A higher fossil capital requirement reduces the weight of the deposit financing wedge Ξ in the loan pricing condition. Intuitively, this depresses fossil investment by making the second purpose of physical investment less valuable, which we can see formally from differentiating $\frac{\partial q^f}{\partial k^f} = \big((1-\kappa^f)\Xi+\beta\big) f(\overline{m}^f)\frac{\overline{m}^f}{k^f} > 0$. To benchmark this effect against emission taxes, we plug the expressions for $\widetilde{p}^f$ and $\frac{\partial q^f}{\partial k^f}$ into equation (21):

$$\psi^f \approx \widetilde{\beta}\Big\{\underbrace{\Big((1-\kappa^f)\Xi + \beta\Big) f(\overline{m}^f)(\overline{m}^f)^2}_{\text{Tilted Regulation + Fin. Frictions}} + 1\Big\}\Big(p^f - \underbrace{\tau^e(1-a) - \frac{\theta_1}{\theta_2+1}a^{\theta_2+1}}_{\text{Tax and Abatement Cost}}\Big) \ . \tag{22}$$

The analytical steps are relegated to Appendix B. Equation (22) shows that, irrespective of financial frictions in the production sector, a higher emission tax directly reduces the expected payoff from fossil energy investment. By contrast, fossil capital requirements indirectly affect fossil energy investment through changes in the loan price schedule. The effectiveness of fossil capital requirements depends on i) the size of the deposit financing wedge and ii) on the relevance of loan financing conditions for optimal investment.

As we show in Appendix B, we can approximate the deposit financing wedge as $\Xi \approx r^{rf} - r^D + F(\overline{\mu})$, such that setting κ^f=100% raises the loan price ceteris paribus by the sum of the bank failure rate (60pbs) and the liquidity premium (40bps). While this adds up to almost 100 basis points, it does not translate into a 100bps increase in the expected payoff from fossil investment on the RHS of equation (22), which would be quite substantial. With corporate default risk, the impact of the deposit financing wedge on investment is reduced by the factor $f(\overline{m}^f)(\overline{m}^f)^2$ in equation (22). This term is related to the second purpose of physical investment and measures by how much higher investment contributes to current dividends through the creation of debt capacity. Specifically, the term $f(\overline{m}^f)\overline{m}^f$ reflects the marginal reduction in default risk by increasing investment and $\overline{m}^f$ is related to the loan financed share of capital. The effect would be largest if firms were completely loan financed ($\overline{m}^f$=1), while fossil capital requirements would have no effect at all if firms were completely equity financed ($\overline{m}^f$=0).

Our baseline calibration implies $f(\overline{m}^f)(\overline{m}^f)^2 \approx 0.135$, such that setting κ^f=100% elevates the long run cost of capital in the fossil sector by around $(1 - 0.08) \cdot 0.135 \cdot$ 100bps $\approx$ 10bps. This is illustrated in the upper left panel of Figure 3. Contrasting this with the effects of emission taxes in the upper right panel of Figure 3, it stands out that emission taxes are two orders of magnitude more powerful in reducing emissions. Finally, in Appendix B, we demonstrate that higher fossil capital requirements reduce the equilibrium fossil energy share and how it distorts the risk-taking decision across sectors.

The generally modest long run effect of bank capital requirements on loan rates, loan volumes, investment and GDP is consistent with similar models, see for example Mendicino et al. (2024) or Angelini et al. (2011) and Kashyap et al. (2010) for earlier references. In Section D.3, we demonstrate that these effects are also consistent with the empirical literature, which corroborates the external validity of our quantitative analysis.

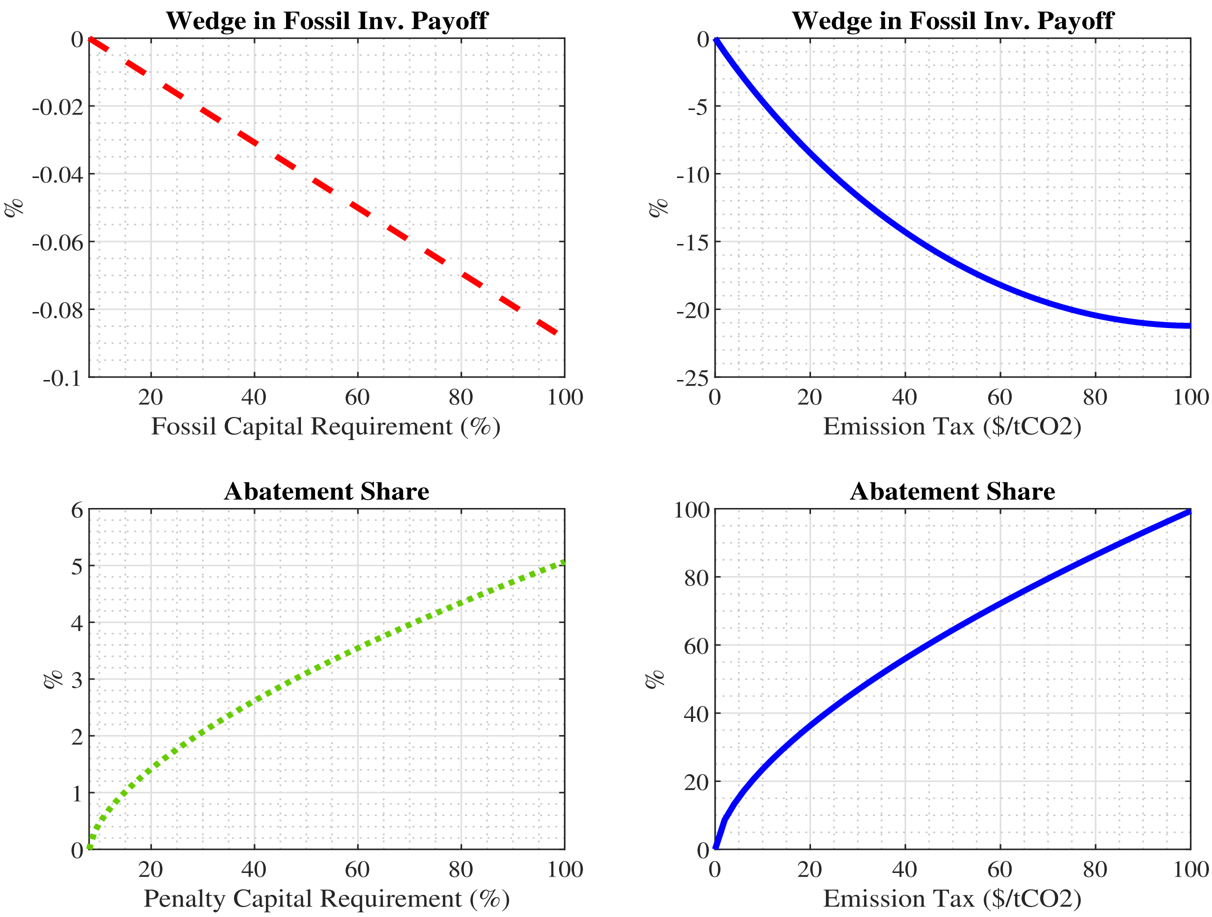

Figure 3: Tilted Bank Capital Requirements versus Emission Taxes. The top row displays the wedge in the expected payoff from fossil investment (equation 22) for different fossil capital requirements (top left) and for different emission taxes (top right). The bottom row displays the share of abated emissions (equation 23) for different penalty requirements κ^{pen}, while fixing κ^{reward}=8% and emission taxes at zero (bottom left) and for different emission taxes (bottom right).

Sustainability Linked Capital Requirements. The transmission mechanism of the SLC to emissions depends on its effect on the optimal abatement effort. We set $\tau^e = 0$ and plug the simplified loan price schedule into (11) to obtain

$$a^* = \left(\frac{(\kappa^{pen} - \kappa^{reward})\Xi\big(1 - F(\overline{m}^f)\big)\frac{l^f}{z^f}}{\theta_1} \right)^{1/\theta_2} . \tag{23}$$

The abatement effort increases in the stringency of the SLC as measured by the $\kappa^{pen} - \kappa^{reward}$ and on the deposit financing wedge Ξ, which determines the relevance of capital requirements for the loan price. The optimal abatement effort also depends on the expression $\big(1 - F(\overline{m}^f)\big)\frac{l^f}{z^f}$, which is linked to financial frictions in the production sector and is strictly smaller than one. Specifically, abatement increases with the relevance of debt financing conditions for firm dividends, captured by the ratio $\frac{l^f}{z^f}$, and decreases in default risk, as this reduces the loan price and hence the relevance of the SLC for current dividends. This is in sharp contrast to the transmission of emission taxes, which is not directly affected by financial frictions in the banking and production sectors.

How much additional abatement effort does the SLC induce in comparison to emission taxes? The bottom left panel of Figure 3 shows the results from varying κ^{pen} while keeping κ^{reward} at 8% and the emission tax at zero. Similar to the full model (Table 2), the abatement share is about 5% for the most restrictive policy, while even 5\$/tCO2 global average emission tax currently in place induces a higher abatement share, as the lower right panel of Figure 3 illustrates. The simplified model reveals that the quantitatively small effect of tilted capital requirements stems from financial frictions that dampen their transmission to the real sector, while the effectiveness of emission taxes is independent from financial frictions.

6 Optimal Bank Capital Requirements

Having discussed the limited efficacy of bank capital requirements as climate policy instrument, we now turn to optimal bank capital requirements and discuss how they depend on climate policy. In our model, optimal bank capital requirements balance a key trade-off between reducing the bank failure rate and maintaining a high supply of deposits to households.

Emission Taxes. We first discuss the effect of emission taxes on macro-financial variables. While such taxes reduce emissions by design, they also depress investment and output in the fossil energy sector. Since the different intermediate goods are imperfect substitutes in the final good production function, this translates into a decline in aggregate investment. Importantly, emission taxes have no (long run) effect on bank failure, since bank leverage is determined by the capital requirement in the long run. Moreover, emission taxes have no direct (long run) effect on firms' default risk, since emission taxes do not affect the benefits of debt over equity financing. Thus, the decline in aggregate investment goes hand in hand with a decline in aggregate loan demand and deposit supply, which reflects the deposit channel of bank capital requirements. Appendix D.2 provides further quantitative details for the outlined effects.

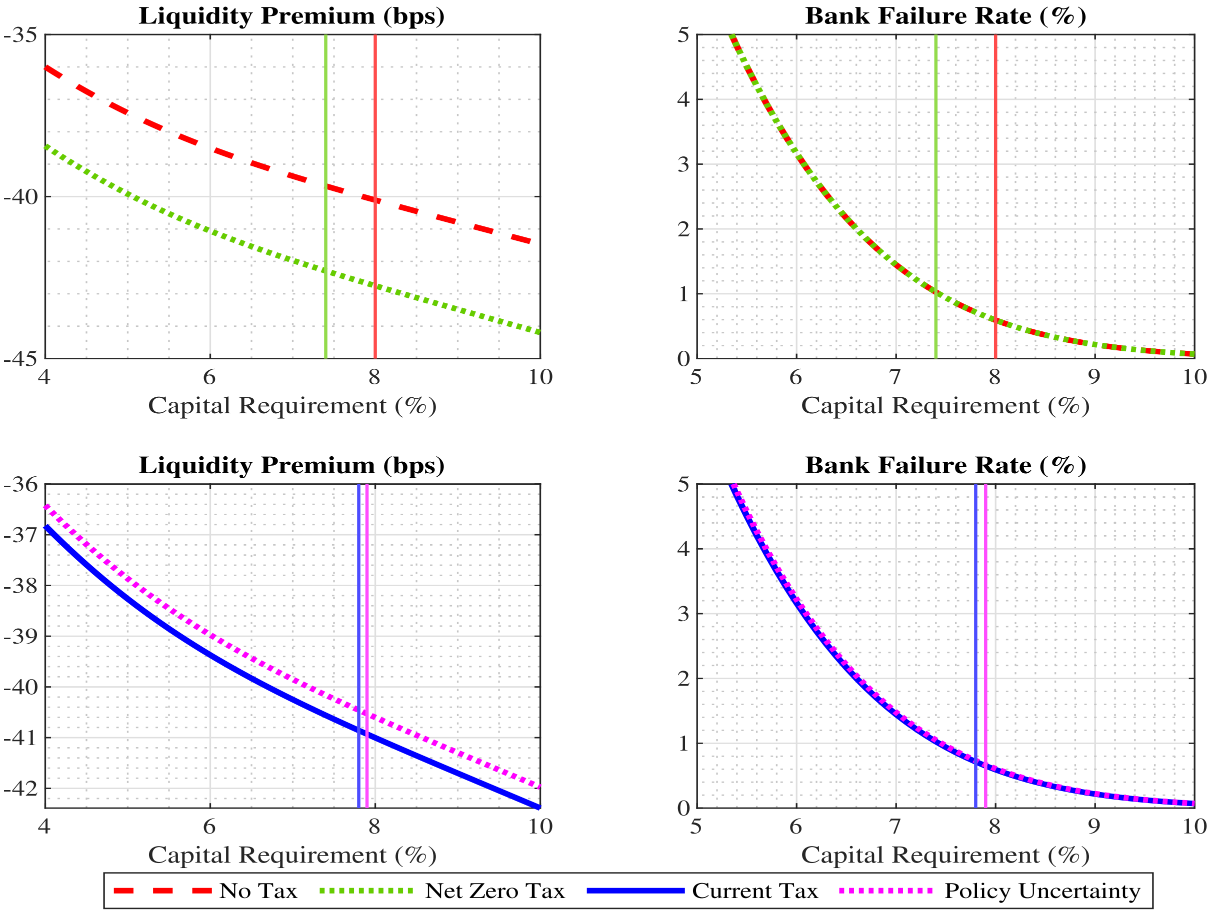

Figure 4: Long Run Effects of Symmetric Capital Requirements. This figure displays the effects of long run capital requirements on the liquidity premium $r^D - r^{rf}$ (in basis points) and the bank failure rate $F(\overline{\mu})$ (in %). The top row compares the baseline case without policies (dashed red line) to net zero taxes (dotted green line). The bottom row illustrates the role of policy uncertainty (dotted magenta line) against constant emission taxes fixed at their current global average of 5\$/tCO2. The vertical lines represent the optimal capital requirement under each climate policy specification.

How do the macroeconomic effects of emission taxes affect optimal bank capital regulation? The upper panels of Figure 4 shows how the symmetric capital requirement κ^{sym} transmits to macroeconomic aggregates without emission taxes (dashed red line) and with net zero emission taxes (dotted green line). The right panel demonstrates by how much tighter requirements reduce the failure probability of banks. At the same time, the liquidity premium widens in the capital requirement due to the deposit scarcity effects discussed above (left panel). This represents the key trade-off for bank capital regulation in the long run. Importantly, the transmission of capital requirements on these variables does not depend on the climate policy regime in place. Rather than affecting the transmission of bank capital requirements to the macroeconomy itself, higher emission taxes affect how the trade-off is resolved optimally. While they are detrimental to liquidity provision they leave the long run bank failure rate unchanged.

As the vertical lines in Figure 4 show, it is optimal to decrease the (symmetric) capital requirement from 8% to 7.4%, which counteracts the adverse effect of emission taxes on liquidity provision. Furthermore, capital requirements affect banks' loan pricing, such that lowering them has a direct welfare-enhancing effect on investment and a welfare-diminishing effect on risk-taking in the non-financial sector, which arise because firms endogenously respond to changes in loan financing conditions. Appendix D.3 shows this in greater detail.

Climate Policy Uncertainty. Up to this point, we have compared macro-financial variables under different emission tax levels, i.e. the climate policy *stance*. In practice, emission taxes are characterized by considerable policy uncertainty. While emission tax shocks have negligible effects on total loans and hence the liquidity premium (lower left panel of Figure 4), they raise the average corporate default and bank failure rates (lower right panel of Figure 4). Intuitively, fossil default rates increase by more in response to a tax hike than they decline in response to a tax cut. It is hence optimal to slightly *raise* the bank capital requirement by 0.1 percentage point. For additional details, we again refer to Appendix D.3.

7 Robustness

In this section, we show that our results are robust to changing parameters in the climate block, modifying the sectoral shares and elasticities in the production sector, and extending the financial sector along different dimensions. In Table 3, we display several key metrics summarizing the implications of tilted and optimal bank capital regulation. Further details are provided in Appendix E. Columns (1) and (2) show the effects of the simple fossil capital requirements on emissions and welfare, while columns (3) and

(4) refer to sustainability linked capital requirements. Column (5) reports the change in optimal capital requirements under net zero taxes relative to the case without climate policy. Column (6) displays the change in optimal capital requirements with climate policy uncertainty, fixing the tax level the current global average of 5\$/tCO2.

	κ^f=100%		SLC		Opt. Adjust. of κ(%)	
	ΔE(%)	ΔV(%)	ΔE(%)	ΔV(%)	Net zero tax	Tax shocks
Baseline Calibration	-0.96	-0.14	-4.59	+0.38	-0.6	+0.1
Panel A: Household and Climate Parameters						
High Abatement Cost	-0.96	-0.13	-3.31	+0.20	-0.6	+0.0
Abatement Investment	-0.97	-0.14	-4.53	+0.36	-0.6	+0.0
High Emission Damages	-0.92	-0.02	-4.41	+0.96	-0.8	+0.1
High Deposit Valuation	-1.50	-0.21	-6.16	+0.46	-0.8	+0.05
Panel B: Sectoral Structure						
Low Energy Share $\widetilde{\nu}$	-0.94	-0.07	-4.38	+0.25	-0.5	+0.05
High Energy-Non-Energy EoS $\widetilde{\epsilon}$	-1.11	-0.11	-4.65	+0.36	-0.5	+0.1
Lower Fossil Share ν	-1.37	-0.06	-4.88	+0.38	-0.6	+0.05
Higher Clean-Fossil EoS ϵ	-1.71	-0.03	-5.27	+0.47	-0.6	+0.15
Panel C: Financial Sector						
High Restructuring Cost	-0.98	-0.14	-4.65	+0.38	-0.6	+0.1
Bond Financing	-0.59	+0.06	-0.60	+0.07	-0.3	+0.1
Heterogeneous Banks	-0.95	-0.14	-4.61	+0.37	-0.7	+0.1
Risk-Free Reserve Asset	-1.13	-0.20	-4.28	+0.25	-0.2	+0.1

Table 3: Robustness. The first columns refer a 100% capital requirement on fossil loans and the middle columns to sustainability linked capital requirements, setting the emission tax to zero, respectively. Column (5) shows how the optimal capital requirement changes if emission taxes increase from zero to the full abatement level. Column (6) shows how the optimal capital requirement changes in the presence of emission tax shocks, fixing the emission tax level at 5\$/tCO2. Welfare is defined in equation (A.31), computed relative to the baseline without climate policies and expressed in consumption equivalents.

Household and Climate Parameters. Motivated by the considerable uncertainty surrounding the severity of climate damages and the cost of abating emissions, we show how the interplay between climate policy and bank regulation depends on several important parameters. In the first row of Table 3, Panel A, we compare our baseline calibration to a higher abatement cost parameter, such that an emission tax of 200\$/tCO2 is required to achieve net zero. This hardly affects the transmission of simple fossil capital requirements but reduces the effectiveness of the SLC, as emission abatement is generally more costly. By contrast, optimal bank capital regulation is hardly affected. The second row of Table 3, Panel A, presents the results from treating abatement as a one-period-ahead investment choice rather than a contemporaneous effort choice like in the baseline model, where firms can freely adjust the share of abatement emissions every period. In this model version, abatement interacts non-trivially with financial frictions, as we discuss in Appendix E.1. It turns out that this model modification has negligible effects on bank capital regulation.

In the third row of Table 3, Panel A, we raise the climate damage parameter so that long run damages associated with unmitigated global warming amount to 8% of GDP, compared to 4% in the baseline calibration. The efficacy of tilted capital requirements is very similar to the baseline, but emission reductions deliver larger welfare gains, such that welfare is almost unaffected by setting κ^f=100% and the SLC raises welfare by almost one percentage point in consumption equivalents. We also consider a larger weight on deposits in the utility function, such that it implies a 60bps liquidity premium and a correspondingly larger deposit financing wedge in the loan price schedule. Hence the effects of tilted bank regulation on emissions and the adverse side effects on liquidity provision are much larger in this case, such that setting κ^f=100% inflicts larger welfare losses, see the fourth row of Table 3, Panel A. Interestingly, the welfare gain of the SLC is slightly larger, as a larger abatement effort has a stronger effect on firms' cost of debt and, thus, dividends in this case. Notably, the optimal adjustment of κ^{sym} in response to net zero taxes and to climate policy uncertainty is similar to the baseline model in both cases. Additional details on these robustness checks are presented in Appendix E.1.

Sectoral Structure. The next set of robustness checks concerns the sectoral structure of the production side. While the sectoral shares are straightforward to measure, they are quite heterogeneous across countries, warranting further robustness checks. Our baseline calibration is based on purposefully conservative parameters, i.e. low elasticities and large weights of the fossil energy sector, as this provides a reasonable upper bound for the adverse medium run consequences of climate policy on liquidity provision. We relax these conservative assumptions one at a time.

First, we consider an energy share $\widetilde{\nu}$ in the production function of 7%, which is used in comparable quantitative work (Golosov et al., 2014) and considerably lower than our baseline value of 10%. Second, we increase the substitution elasticity between energy and non-energy goods to $\widetilde{\epsilon}$=0.4, as in Coenen et al. (2024). Third, we set ν such that only 70% of firms initially operate the fossil technology, compared to 80% in the baseline calibration. Such lower values have been used in calibrations of comparable DSGE models with dis-aggregated energy sectors, see the discussion in Appendix E.2. Fourth, we set substitution elasticity within the energy sector to ϵ=5, following the argument of Acemoglu et al. (2012), who consider our baseline parameter choice of ϵ=3 as a lower bound for the longer run and consider a range of even higher values.

Changing the sectoral structure does not affect the transmission of tilted bank capital regulation per se, as it does not change the wedges in the first-order conditions for capital and abatement, respectively. However, Panel B of Table 3 demonstrates that the relevance of these wedges for aggregate emissions and the welfare gain of climate policy are affected. This follows from the smaller macroeconomic relevance of fossil energy firms, either simply through their smaller weight or through enhanced substitutability. Bank regulation has stronger effects on emissions, while the adverse side effects on liquidity provision are less pronounced. Consequently, the welfare gains of tilted bank regulation are slightly larger than in the baseline model. Similarly, the optimal response of bank regulation to the emission tax level and to policy uncertainty hardly changes as well. We provide further details on these robustness checks in Appendix E.2.

Financial Frictions and Model Extensions. Our baseline model is calibrated to US data, which exhibit arguably very similar patterns as other advanced economy data, see Appendix D, but might not be representative of financial frictions on a global level. In particular, it is reasonable to assume that financial intermediation is less effective in emerging economies. The first row of Panel C (Table 3) demonstrates that our results are robust to increasing the cost of debt restructuring φ. Drawing from a recent literature on optimal bank capital regulation in the presence of non-bank lending (Mishin, 2023; Dempsey, 2024; Begenau and Landvoigt, 2021), we also consider an extension where fossil energy firms can tap the corporate bond market as alternative funding source, see Appendix E.3 for the analytical steps and parameterization in this extension. In this case, tilted bank capital requirements have a much smaller effect on emissions, but also less pronounced side effects on liquidity provision. Consequently, welfare increases even for κ^f=100%, while the SLC is much less beneficial.

Next, we relax the assumption of perfect diversification of banks across sectors. However, even assuming that banks only lend to one sector neither affects the transmission of tilted bank capital regulation or the optimal capital requirement in response to emission taxes, see the third row of Table 3, Panel C. In the last row of Table 3, we allow banks to hold government bonds as a nominally risk free asset, which are a large item on bank balance sheets in practice. We increase households' valuation of liquidity services to match the deposit spread, see Appendix E.3 for further details. Hence, the welfare gains of tilted capital requirements are smaller than in the baseline. At the same time, emission taxes inflict smaller side effects on liquidity provision and the optimal capital requirement declines by merely 0.2 percentage points.

8 Transition Dynamics

Lastly, we compute the macroeconomic and climate impact of tilted capital requirements when they support the net zero transition. We solve and simulate the model under perfect foresight. We interpret 2025 as the start year and assume that emission taxes increase linearly from 5\$/tCO2 to the full abatement level of 100\$/tCO2 over 25 years, i.e. in 2050. This path, reflected by the solid blue line in Figure 5, is consistent with climate policy goals set out in the Paris agreement. The lower left panel shows that atmospheric carbon dioxide levels are slightly smaller if sustainability linked capital requirements are used alongside emission taxes (dotted green line). The difference is quite small and translates into a negligible welfare gain.

The reason behind this at first glance puzzling result is that the SLC quickly loses its relevance. This can be seen directly from the optimal abatement effort a_t^* in equation (11). Due to the convexity of the abatement cost function, the additional abatement effort induced by the SLC declines in the emission tax. At the same time, the SLC lowers

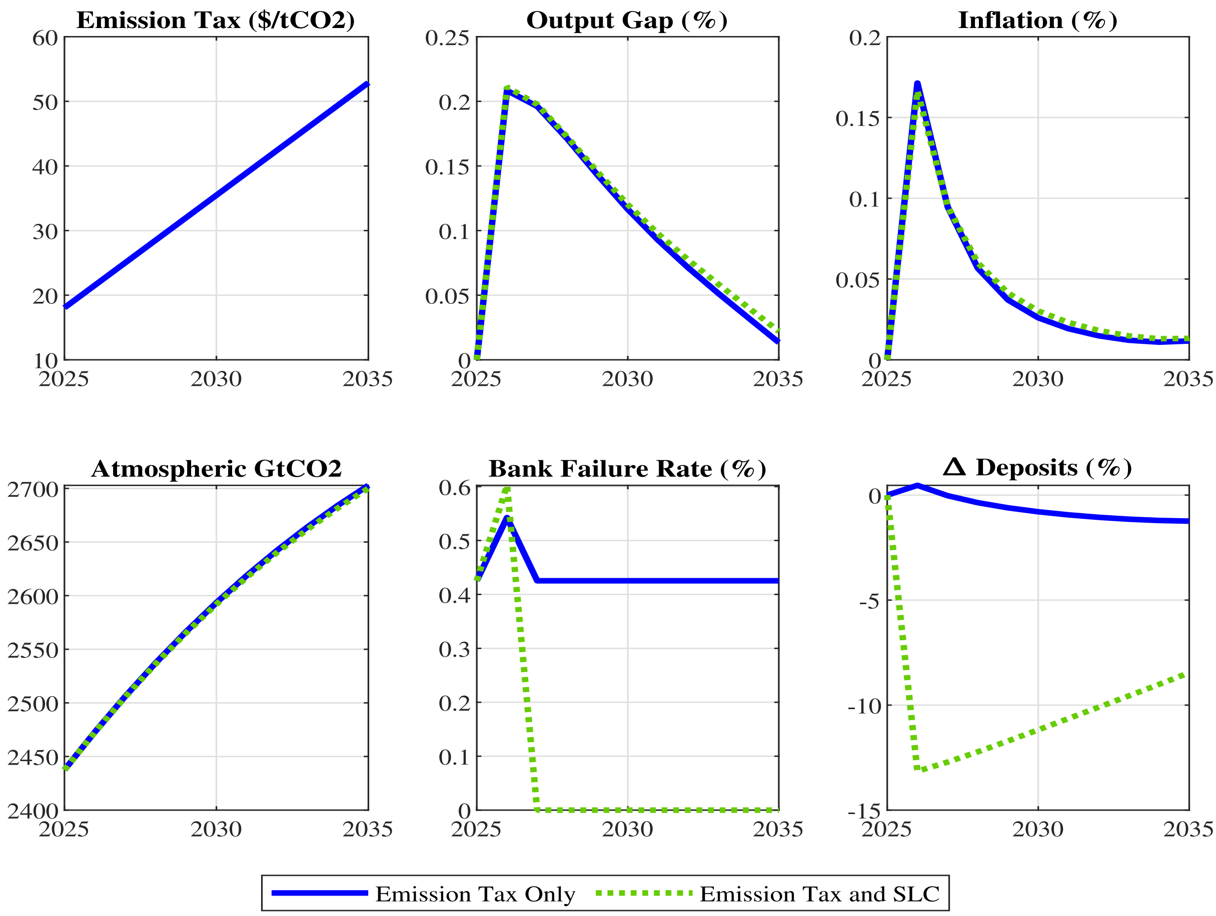

Figure 5: Transition Path. The linear emission tax path from 5$/tCO2 in 2025 to 100$/tCO2 in 2050 is reflected by the solid blue line. Adding sustainability linked capital requirements to the policy mix is reflected by the dotted green line.

the bank failure rate and deposit supply, which is again associated with adverse side effects through the deposit channel of bank regulation. We discuss the sectoral effects and aggregate consequences of the net zero transition in Appendix D.4.

Finally, we also discuss whether and how much tilted bank capital requirements soften the short term trade-offs faced by policymakers when the economy embarks on an ambitious climate policy path. The top row of Figure 5 shows that the transition induces a spike in the output gap and inflation, consistent with recent empirical evidence (Hensel et al., 2024). In our model, this follows from a fossil energy supply contraction due to higher spending on emission taxes and abatement. Final good producers cannot perfectly switch to clean or non-energy inputs so that marginal costs rise. Nominal rigidities in final good production and on the labor market imply that prices and wages cannot increase fast enough in order to reflect these higher marginal costs. Thus, output is temporarily above its natural level, consistent with a negative supply shock.

The SLC slightly counteracts the marginal cost increase by stimulating loan take-up. Importantly, this is due to a general equilibrium effect stemming from the scarcity of deposits that we already described in Table 2. This slightly stimulates investment and output in the intermediate sectors, which reduces marginal costs for final goods producers, thereby relieving inflationary pressures. As this effect is comparatively small, it does not materially affect the basic trade-off between reducing emissions and impairing the supply of liquidity that tilted bank capital regulation faces in our model.

9 Conclusion

In this paper, we have proposed and calibrated a multi-sector DSGE model with two layers of default to study the interactions between bank capital regulation and climate policy. We find that simple fossil capital requirements have a negligible effect on emissions that are accompanied by negative liquidity supply effects, such that they reduce welfare. Sustainability linked capital requirements reduce emissions more effectively and have smaller side effects, but the welfare gain is still smaller than a modest emission tax. If climate policy is implemented using emission taxes, it is optimal to relax capital requirements symmetrically across sectors to counter negative loan demand and deposit supply effects of emission taxes, while optimal capital requirements are slightly larger in the presence of climate policy uncertainty. Furthermore, tilted bank capital requirements do not soften short run trade-offs between inflation and output contractions during the net zero transition. These results are robust to reasonable variations in the parameters governing climate policy and the sectoral structure of the economy. Lastly, it should be noted that our model abstracts from endogenous technological change in the cost of abating emissions, which can give rise to multiplicity in balanced growth paths (Campiglio et al., 2024b). The very small effects of tilted bank capital regulation suggests that financial policies are probably not decisive in selecting a low emission growth path.

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Online Appendix to "Climate Change and the Macroeconomics of Bank Capital Regulation"

Francesco Giovanardi and Matthias Kaldorf

A Model Appendix

This section provides additional details on households' labor supply in A.1. Section A.2 contains analytical steps to derive banks' loan pricing condition that links firms' risk choice to a loan price. We then derive intermediate good firms' loan demand and investment in Section Section A.3. Lastly, we derive investment good supply in Section A.4 and derive the New Keynesian Phillips curve in Section A.5.

A.1 Households: Wage Setting

The wage setting problem of households closely follow the exposition in Smets and Wouters (2003). There is a continuum of households supplying differentiated labor types in a monopolistically competitive way. A fraction Ψ_N of households cannot adjust wages each period, while the fraction $1-\Psi_N$ that is allowed to adjust the wage sets it to maximize lifetime utility, subject to the labor supply function $n_t(i) = \left(\frac{W_t(i)}{W_t}\right)^{-\widetilde{\phi}} n_t$. Here, $\widetilde{\phi}$ is the demand elasticity between labor types, n_t is aggregate labor demand and W_t is the aggregate nominal wage:

$$W_t = \left(\int \left(W_t(i)\right)^{1-\widetilde{\phi}} di \right)^{\frac{1}{1-\widetilde{\phi}}} \tag{A.24}$$

Solving the maximization problem of the household for the optimal wage yields the following wage markup equation:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} (\beta\Psi_N)^t \frac{W_t^*}{P_t} \frac{n_t c_t^{-\gamma_C}}{\pi_t} \frac{\widetilde{\phi}-1}{\widetilde{\phi}} (1+\tau^N) = \mathbb{E}_0 \sum_{t=0}^{\infty} (\beta\Psi_N)^t \omega_N n_t^{1+\gamma_N} . \tag{A.25}$$

We assume that labor is subsidized to eliminate the steady state wage markup. For the limiting case of flexible prices, the real wage would satisfy $w_t c_t^{-\gamma_C} = \omega_N n_t^{\gamma_N}$. Equation (A.25) can be written recursively as

$$\overline{w}_t^{1+\widetilde{\phi}\gamma_N} X_{1,t} = X_{2,t} \tag{A.26}$$

where the auxiliary variables $X_{1,t}$ and $X_{2,t}$ are defined as follows:

$$X_{1,t} = w_t^{\widetilde{\phi}} n_t c_t^{-\gamma_C} + \beta\Psi_N \pi_{t+1}^{\widetilde{\phi}-1} X_{1,t+1} , \tag{A.27}$$

$$X_{2,t} = w_t^{\widetilde{\phi}(1+\gamma_N)} \omega_N n_t^{1+\gamma_N} + \beta\Psi_N \pi_{t+1}^{\widetilde{\phi}(1+\gamma_N)} X_{2,t+1} . \tag{A.28}$$

The law of motion for aggregate real wages can hence be written as:

$$w_t^{1-\widetilde{\phi}} = \Psi_N \left(\frac{w_{t-1}}{\pi_t}\right)^{1-\widetilde{\phi}} + (1-\Psi_N)\big(w_t^*\big)^{1-\widetilde{\phi}} . \tag{A.29}$$

and the aggregate labor supply index becomes:

$$\mathcal{N}_t = \Delta_t n_t , \tag{A.30}$$

where $\Delta_t = \int_0^1 (W_t(i)/W_t)^{-\widetilde{\phi}} di$ is an index of wage dispersion and n_t is labor demand. As in Gali and Monacelli (2016), we can define the welfare-relevant measure of wage dispersion $\widetilde{\Delta}_t$ as

$$\widetilde{\Delta}_t^{1+\gamma_N} = \int_0^1 (W_t(i)/W_t)^{-\widetilde{\phi}(1+\gamma_N)} di ,$$

which can be written recursively as:

$$\widetilde{\Delta}_t^{1+\gamma_N} = (1-\Psi_N)\left(\frac{\overline{w}_t}{w_t}\right)^{-\widetilde{\phi}(1+\gamma_N)} + \Psi_N\left(\frac{w_t\pi_t}{w_{t-1}}\right)^{\widetilde{\phi}(1+\gamma_N)}\widetilde{\Delta}_{t-1}^{1+\gamma_N} .$$

The welfare function then becomes

$$V_0 = \mathbb{E}_0\sum_{t=0}^{\infty}\beta^t\Big(\frac{c_t^{1-\gamma_C}}{1-\gamma_C} - \omega_N\frac{\widetilde{\mathcal{N}}_t^{1+\gamma_N}}{1+\gamma_N} + \omega_D\frac{d_{t+1}^{1-\gamma_D}}{1-\gamma_D}\Big) , \tag{A.31}$$

where $\widetilde{\mathcal{N}}_t = n_t\widetilde{\Delta}_t^{1+\gamma_N}$.

A.2 Banks

We first show that banks' optimal funding choice implies that the bank capital requirement is always binding. As in Mendicino et al. (2024), banks are atomistic and owned by households who are perfectly diversified across all banks. Since banks are atomistic, an individual banks' dividends do not affect aggregate consumption. Hence, banks treat the sdf of their owner as exogenously given and behave *as if* they were risk neutral. This allows us to abstract from precautionary motives in banks' shareholder value maximization problem, which are for example studied in Corbae and D'Erasmo (2021) or Jamilov and Monacelli (2025).

The presence of idiosyncratic bank risk shocks implies that banks are heterogeneous *ex post* and some of them fail. However, we can exploit that bank productivity shocks are i.i.d. and that failing banks are restructured immediately, which - together with the absence of precautionary motives - ensures that banks are identical at the end of each period when they make the loan pricing decision.[6] The relevant part of the shareholder

[6] Alternatively, such a bailout guarantee can be expressed as state-dependent transfer. Bank dividends are then given by

$$div_t^b = \mu_t\sum_s \mathcal{R}_t^s l_t^s - (1+r_{t-1}^D)\frac{d_t}{\pi_t} + S_t^{bail} + d_{t+1} - \sum_s q(\overline{m}_{t+1}^s)l_{t+1}^s .$$

If the transfer

$$S_t^{bail} = \mathbb{1}\{\mu_t < \overline{\mu}_t\}\big((1+r_{t-1}^D)\frac{d_t}{\pi_t} - \mu_t\sum_s \mathcal{R}_t^s l_t^s\big)$$

exactly covers the shortfall, period t dividends are identical to the formulation using a transfer of bank value maximization problem of the representative bank is given by

$$\max_{d_{t+1},\{l^s_{t+1}\}} d_{t+1} - \sum_s q^s_t l^s_{t+1} + \mathbb{E}_t \Lambda_{t,t+1} \int_{\overline{\mu}_{t+1}}^{\infty} \mu_{t+1} \sum_s \mathcal{R}^s_{t+1} l^s_{t+1} - (1+r^D_t)\frac{d_{t+1}}{\pi_{t+1}} dF(\mu_{t+1}) \; .$$

Differentiating with respect to d_{t+1} reveals that raising deposits increases bank dividends in period t by one unit. The expected discounted repayment obligations in period $t+1$ amount to $\mathbb{E}_t \Lambda_{t,t+1} \frac{1+r^D_t}{\pi_{t+1}} \big(1 - F(\overline{\mu}_{t+1})\big)$. Deposit financing is cheaper than equity financing for two reasons. First, the liquidity services of deposits make them relatively cheap and, second, the risk of bank failure combined with the bailout guarantee implies that raising one unit of deposits in period t only involves an expected real repayment of $1 - F(\overline{\mu}_{t+1})$ units in $t+1$. Together with banks' risk neutrality, this ensures that the capital requirement binds in all states.

Loan Pricing. Following Gertler and Kiyotaki (2010), equity can be interpreted as a transfer from households to bankers, while dividends are a transfer from bankers to households. Bank equity at the end of each period is implicitly defined by the accounting identity: $(1+r^D_t)\mathbb{E}_t \frac{d_{t+1}}{\pi_{t+1}} + \eta_{t+1} = \sum_s \mathbb{E}_t \mathcal{R}^s_{t+1} l^s_{t+1}$, where and where we have used that $\mathbb{E}_t \mu_{t+1} = 1$. To derive the break-even condition on loan prices, we plug the (binding) bank equity constraint (4) into the maximization problem:

$$\begin{aligned}\max_{\{l^s_{t+1}\}} \quad & \frac{\pi_{t+1}}{1+r^D_t} \sum_s (1-\kappa^s_t)\mathbb{E}_t \mathcal{R}^s_{t+1} l^s_{t+1} - \sum_s q^s_t l^s_{t+1} \\ & + \mathbb{E}_t \Lambda_{t,t+1} \int_{\overline{\mu}_{t+1}}^{\infty} \mu_{t+1} \sum_s \mathcal{R}^s_{t+1} l^s_{t+1} - (1-\kappa^s_t)\mathcal{R}^s_{t+1} l^s_{t+1} dF(\mu_{t+1}) \; ,\end{aligned}$$

and differentiate w.r.t. l^s_{t+1} to get

$$0 = \frac{\pi_{t+1}(1-\kappa^s_t)\mathbb{E}_t \mathcal{R}^s_{t+1}}{1+r^D_t} - q^s_t + \mathbb{E}_t \Lambda_{t,t+1} \Big((1 - G(\overline{\mu}_{t+1}))\mathcal{R}^s_{t+1} - (1 - F(\overline{\mu}_{t+1}))(1-\kappa^s_t)\mathcal{R}^s_{t+1} \Big) \; .$$

Rearranging for q^s_t yields the loan pricing condition (5):

$$q^s_t = \mathbb{E}_t \Big((1-\kappa^s_t)\Xi_{t+1} + \overline{\Lambda}_{t,t+1} \Big) \mathcal{R}^s_{t+1} \; ,$$

assets to the public sector.

where

$$\overline{\Lambda}_{t,t+1} \equiv \Lambda_{t,t+1}\Big(1 - G(\overline{\mu}_{t+1})\Big) \quad \text{and} \quad \Xi_{t+1} \equiv \frac{\pi_{t+1}}{1+r_t^D} - \Lambda_{t,t+1}\big(1 - F(\overline{\mu}_{t+1})\big)$$

Plugging the loan payoff (7) into the loan pricing condition (5), we obtain the loan price schedule that makes the dependency of borrowing conditions on firms' risk choices explicit

$$q(\overline{m}^s_{t+1}) = \mathbb{E}_t\Big((1-\kappa^s_t)\Xi_{t+1}+\overline{\Lambda}_{t,t+1}\Big)\left(\frac{\chi}{\pi_{t+1}}\Big(1-F(\overline{m}^s_{t+1})+\frac{G(\overline{m}^s_{t+1})}{\overline{m}^s_{t+1}}-\varphi F(\overline{m}^s_{t+1})\Big)+(1-\chi)q(\overline{m}^s_{t+2})\right) , \tag{A.32}$$

and enters the maximization problem of intermediate good firms as an additional constraint. The derivative of the loan price schedule with respect to the risk choice $\overline{m}^s_{t+1}$ follows as:

$$\frac{\partial q^s_t}{\partial \overline{m}^s_{t+1}} = \mathbb{E}_t\Big((1-\kappa^s_t)\Xi_{t+1}+\overline{\Lambda}_{t,t+1}\Big)\cdot\Big(-\frac{\chi G(\overline{m}^s_{t+1})}{\pi_{t+1}(\overline{m}^s_{t+1})^2} - \frac{\chi\varphi F'(\overline{m}^s_{t+1})}{\pi_{t+1}} + (1-\chi)\frac{\partial q^s_{t+1}}{\partial \overline{m}^s_{t+2}}\frac{\partial \overline{m}^s_{t+2}}{\partial \overline{m}^s_{t+1}}\Big) . \tag{A.33}$$

When the SLC is active, plugging the linear functional form (9) and the loan payoff (7) into the loan pricing condition (5), we obtain the extended loan price schedule that depends on the abatement effort:

$$q(\overline{m}^f_{t+1}, a_t) = \mathbb{E}_t\Big(\big\{1 - \kappa^{pen}_t + a_t(\kappa^{pen}_t - \kappa^{reward}_t)\big\}\Xi_{t+1} + \overline{\Lambda}_{t,t+1}\Big)\cdot \mathcal{R}^f_{t+1} . \tag{A.34}$$

This is only relevant for fossil energy firms and has an entirely analogous derivative with respect to the risk choice as the baseline loan price schedule. The first-order condition with respect to the abatement effort reads:

$$\frac{\partial q^f_t}{\partial a_t} = (\kappa^{pen}_t - \kappa^{reward}_t)\mathbb{E}_t\Xi_{t+1}\mathcal{R}^f_{t+1} . \tag{A.35}$$

Additional Variables. The cyclical properties of the deposit mark-down are an important untargeted moment that further corroborates our model's external validity. We have shown in the main text (Table 1) that the mark-down is counter-cyclical, i.e. it widens during a recession. The mark-down is defined as:

$$markdown_t \equiv \frac{1+r_t^D}{1+r_t^{rf}} \,. \tag{A.36}$$

Combining equation (1) with (2), the mark-down simplifies to $1-\omega_D \frac{d_{t+1}^{-\gamma_D}}{c_t^{-\gamma_C}}$. The negative markup cyclicality is therefore directly associated with the expression $\omega_D \frac{d_{t+1}^{-\gamma_D}}{c_t^{-\gamma_C}}$ and hence with the marginal utility of consumption and marginal liquidity benefit of deposits. During a recession, deposits and consumption both decline. Importantly, households' relative risk aversion over consumption γ_C is calibrated to a larger value than the risk aversion over liquidity services γ_D. Consequently, the marginal utility of deposits declines by less than the marginal utility of consumption, such that their ratio increases and the mark-down is counter-cyclical.

This mechanism also arises in Jamilov and Monacelli (2025), Section 6.3, although our model abstracts from bank market power. The comparatively small effect on deposits is reinforced in our setting because leverage and loan demand are "sticky" (Gomes et al., 2016). Hence, the size of bank balance sheets and the quantity of deposits is less responsive to productivity shocks.

A.3 Intermediate Good Firms

The relevant part of the shareholder value maximization problem of a fossil energy firm in period t, which is given by equation (10), reads

$$\begin{aligned}
\max_{k_{t+1}^f, l_{t+1}^f, \overline{m}_{t+1}^f} \; & -\psi_t^f(1+\tau^f)k_{t+1}^f + q(\overline{m}_{t+1}^f, a_t)\Big(l_{t+1}^f - (1-\chi)\frac{l_t^f}{\pi_t}\Big) \\
& + \mathbb{E}_t \widetilde{\Lambda}_{t,t+1}\Bigg(\int_{\overline{m}_{t+1}^f}^{\infty} \widetilde{p}_{t+1}^f m k_{t+1}^f - \chi \frac{l_{t+1}^f}{\pi_{t+1}} dF(m) \\
& \qquad + \psi_{t+1}^f(1+\tau^f)(1-\delta_K)k_{t+1}^f + q(\overline{m}_{t+2}^f, a_{t+1})\Big(l_{t+2}^f - (1-\chi)\frac{l_{t+1}^f}{\pi_{t+1}}\Big)\Bigg) \,.
\end{aligned}$$

The first-order conditions directly follow from differentiating with respect to k_{t+1}^f, l_{t+1}^f and $\overline{m}_{t+1}^f$.

As pointed out by Gomes et al. (2016), characterizing the loan and investment choices in the intermediate good sector needs to take the stickiness of firm leverage into account. Such stickiness arises since debt is long-term and next period's risk choice $\overline{m}^f_{t+2}$ depends on the current risk choice $\overline{m}^f_{t+1}$. This affects both the risk choice (14) and the derivative of the loan price schedule (A.33). Following Gomes et al. (2016), we pin down $\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}}$, which is an object that depends on the unknown policy function for risk choice, by using an additional condition. With a slight abuse of notation, we omit the potential dependency of q^f_t on a_t but note that the same algebra also applies to the sustainability linked capital requirement.

We arrange the first-order condition for loans, equation (12), for the Lagrangian multiplier on the risk choice,

$$\lambda^f_t = \frac{l^f_{t+1}}{\overline{m}^f_{t+1}} q(\overline{m}^f_{t+1}) - \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}} \frac{l^f_{t+1}}{\overline{m}^f_{t+1}} \left(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi) q(\overline{m}^f_{t+2}) \right) , \tag{A.37}$$

and plug this expression into the first-order condition for capital (13) to get:

$$\begin{aligned}(1+\tau^f)\psi^f_t - &\mathbb{E}_t \frac{l^f_{t+1}}{\pi_{t+1} k^f_{t+1}} \left(q(\overline{m}^f_{t+1})\pi_{t+1} - \widetilde{\Lambda}_{t,t+1} \left\{ \chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi) q(\overline{m}^f_{t+2}) \right\} \right) \\ &= \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} \Big((1-\delta_k)\psi^f_{t+1}(1+\tau^f) + (1 - G(\overline{m}^f_{t+1}))\widetilde{p}^f_{t+1} \Big) .\end{aligned}$$

Multiplying both sides by $\frac{\chi}{\widetilde{p}^f_{t+1}}$,

$$\begin{aligned}\frac{\chi}{\widetilde{p}^f_{t+1}} \psi^f_t (1+\tau^f) - &\overline{m}^f_{t+1} \mathbb{E}_t \left(q(\overline{m}^f_{t+1})\pi_{t+1} - \widetilde{\Lambda}_{t,t+1} \left\{ \chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi) q(\overline{m}^f_{t+2}) \right\} \right) \\ &= \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} \left(\frac{\chi}{\widetilde{p}^f_{t+1}} (1-\delta_k)\psi^f_{t+1}(1+\tau^f) + \chi(1 - G(\overline{m}^f_{t+1})) \right) ,\end{aligned}$$

and differentiating this expression w.r.t. $\overline{m}^f_{t+1}$, we obtain

$$\begin{aligned}&- q(\overline{m}^f_{t+1})\pi_{t+1} + \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} \left(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi) q(\overline{m}^f_{t+2}) \right) \\ &- \overline{m}^f_{t+1} \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} \left(\frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \pi_{t+1} + \chi f(\overline{m}^f_{t+1}) + (1-\chi) \frac{\partial q^f_{t+1}}{\partial \overline{m}^f_{t+2}} \frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}} \right) = -\chi \mathbb{E}_t \widetilde{\Lambda}_{t,t+1} g(\overline{m}^f_{t+1}) .\end{aligned}$$

We can simplify this expression by exploiting that the derivative of the conditional mean $g(\overline{m}^f_{t+1})$ satisfies $g(\overline{m}^f_{t+1}) = \overline{m}^f_{t+1} f(\overline{m}^f_{t+1})$:

$$q(\overline{m}^f_{t+1}) + \overline{m}^f_{t+1}\frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} = \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}}\Big(\chi(1-F(\overline{m}^f_{t+1})) + (1-\chi)\Big\{q(\overline{m}^f_{t+2}) + \overline{m}^f_{t+1}\frac{\partial q^f_{t+1}}{\partial \overline{m}^f_{t+2}}\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}}\Big\}\Big)\ , \tag{A.38}$$

Equation (A.38) is an additional equilibrium condition that pins down the slope of the policy function for leverage $\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}}$. Intuitively, its LHS of summarizes the additional dividends today financed by issuing one unit of loans net of a debt dilution term, as $q'(\overline{m}^f_{t+1})$ is negative. The RHS of equation (A.38) indicates that increasing leverage today increases the expected repayment obligation from the maturing loan share χ and also affects how a higher leverage affects the additional dividends tomorrow that can be financed by issuing one unit of loans. This condition is derived entirely analogously for clean and non-energy firms.

Loan Payoff and Recovery Rate. The production revenues that banks seize in the default case are distributed equally among the holders of maturing loans χl^s_t. This enters the expected loan payoff (7) as a *partial* expectation $\int_0^{\overline{m}^s_t} \frac{p^s_t m_t k^s_t}{\chi l^s_t} dF(m_t) = \frac{G(\overline{m}^s_t)}{\overline{m}^s_t}$. These revenues enter the recovery rate as a *conditional* expectation $\mathbb{E}_t[\frac{p^s_t m_t k^s_t}{\chi l^s_t}|m_t < \overline{m}^s_t]$, which relates the average revenues of a defaulter to the promised loan repayment. The conditional expectation obtains from dividing the partial expectation by the default probability, such that we have $\frac{G(\overline{m}^s_t)}{F(\overline{m}^s_t)\overline{m}^s_t}$. Hence the expected recovery rate per unit of loans is given by the realized payoff from holding a distressed loan relative to the promised payoff minus the per-unit restructuring cost φ.

$$recov(\overline{m}^s_t) = \frac{G(\overline{m}^s_t)}{F(\overline{m}^s_t)\overline{m}^s_{t+1}} - \varphi\ . \tag{A.39}$$

The aggregate recovery rate is defined as the loan volume weighted recovery rate in each sector, i.e. $recov_t \equiv \sum recov(\overline{m}^s_t)\frac{l^s_t}{\sum l^s_t}$.

A.4 Investment Good Firms

There is a representative producer for each of the three investment goods that intermediate firms acquire at price ψ_t^s. To produce one unit of each investment good, these firms use $\left(1+\frac{\Psi_I}{2}(\frac{i_t^s}{i_{t-1}^s})\right)$ units of the final good. The profit maximization problem

$$\max_{\{i_t^s\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{-\gamma_C}}{c_0^{-\gamma_C}} \left(\psi_t^s i_t^s - \left(1 + \frac{\Psi_I}{2}(\frac{i_t^s}{i_{t-1}^s} - 1)^2\right) i_t^s \right)$$

yields the first-order condition for (type-specific) investment good supply

$$\psi_t^s = 1 + \frac{\Psi_I}{2}\left(\frac{i_t^s}{i_{t-1}^s} - 1\right)^2 + \Psi_I\left(\frac{i_t^s}{i_{t-1}^s} - 1\right)\frac{i_t^s}{i_{t-1}^s} - \mathbb{E}_t \Lambda_{t,t+1} \Psi_I \left(\frac{i_{t+1}^s}{i_t^s} - 1\right)\left(\frac{i_{t+1}^s}{i_t^s}\right)^2 . \tag{A.40}$$

Intuitively, the marginal effect of an additional unit of loans on current dividends (LHS) has to equal the marginal effect on dividends in the next period (RHS). These are given by the expected repayment on the maturing share χ and the marginal effect of an additional unit of debt issued next period, which applies to the rollover share $(1-\chi)$.

A.5 Final Good Firms

The cost minimization problem of final good firms yields the following standard first-order conditions:

$$mc_t \alpha \widetilde{\nu} \nu \frac{y_t}{\widetilde{z}_t} \left(\frac{\widetilde{z}_t}{z_t^e}\right)^{\frac{1}{\widetilde{\epsilon}}} \left(\frac{z_t^e}{z_t^c}\right)^{\frac{1}{\epsilon}} = p_t^c , \tag{A.41}$$

$$mc_t \alpha \widetilde{\nu} (1-\nu) \frac{y_t}{\widetilde{z}_t} \left(\frac{\widetilde{z}_t}{z_t^e}\right)^{\frac{1}{\widetilde{\epsilon}}} \left(\frac{z_t^e}{z_t^f}\right)^{\frac{1}{\epsilon}} = p_t^f , \tag{A.42}$$

$$mc_t \alpha (1-\widetilde{\nu}) \frac{y_t}{\widetilde{z}_t} \left(\frac{\widetilde{z}_t}{z_t^n}\right)^{\frac{1}{\widetilde{\epsilon}}} = p_t^n , \tag{A.43}$$

$$mc_t (1-\alpha) \frac{y_t}{n_t} = w_t , \tag{A.44}$$

where mc_t is the real marginal cost of production for the final good. Denoting with ϕ the elasticity of substitution across final goods, final good monopolists face price rigidities à la Rotemberg, with Ψ_P being the parameter governing the degree of nominal rigidity.

The price-setting maximization problem of final good producer i is then given by

$$\max_{P_t(i)} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{-\gamma_C}}{c_0^{-\gamma_C}} \left\{ \left(\frac{P_t(i)}{P_t} \right)^{-\phi} \left(\frac{P_t(i)}{P_t} - mc_t \right) y_t - \frac{\Psi_P}{2} \left(\frac{P_t(i)}{P_{t-1}} - 1 \right)^2 y_t \right\} .$$

Solving the maximization problem and imposing symmetry among producers, we arrive at the standard New Keynesian Philips curve,

$$\mathbb{E}_t \Lambda_{t,t+1} \frac{y_{t+1}}{y_t} \left(\pi_{t+1} - 1 \right) \pi_{t+1} + \frac{\phi}{\Psi_P} \left(\mathrm{mc}_t - \frac{\phi - 1}{\phi} \right) = \left(\pi_t - 1 \right) \pi_t \ . \tag{A.45}$$

B Simplified Model

In this section, we provide additional analytical steps for the simplified model presented in Section 5.

Investment and Loan Demand. In the simplified case with one period debt and without nominal or real rigidities, the multiplier on the risk choice by fossil energy firms reduces to $\lambda^f_{t+1} = -l^f_{t+1} q'(\overline{m}^f_{t+1})$ and the default threshold becomes $\overline{m}^f_{t+1} = \frac{l^f_{t+1}}{k^f_{t+1}\widetilde{p}^f_{t+1}}$. Exploiting this together with full capital depreciation in the first order conditions for investment and debt (equations (12) and (13)) yields

$$q(\overline{m}^f_{t+1}) + \frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \overline{m}^f_{t+1} = \widetilde{\beta}\big(1 - F(\overline{m}^f_{t+1})\big) \ , \tag{B.46}$$

$$\psi^f_t - \frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \frac{\partial \overline{m}^f_{t+1}}{\partial k^f_{t+1}} l^f_{t+1} = \widetilde{\beta}\big(1 - G(\overline{m}^f_{t+1})\big)\widetilde{p}^f_{t+1} \ . \tag{B.47}$$

The simplified first-order condition for loans, equation (B.46), requires that the marginal benefit of issuing one unit of debt today, given by the loan price $q(\overline{m}^f_{t+1})$ and net of a debt dilution term $\frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \overline{m}^f_{t+1} < 0$ on the LHS equals its cost, i.e. the discounted expected repayment obligation (RHS).

The simplified first-order condition for capital, equation (B.47), requires that the cost of purchasing one unit of capital (LHS) equals its discounted expected payoff (RHS). The expected payoff is given by next period's fossil energy price net of expenses related to emissions $\widetilde{p}^f_{t+1}$, multiplied by the expected profitability conditional on not defaulting $1 - G(\overline{m}^f_{t+1})$. This term reflects that a higher default probability makes investment into fossil capital less attractive since firm managers benefit from the investment in fewer states, akin to a debt overhang problem. The term $\frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \frac{\partial \overline{m}^f_{t+1}}{\partial k^f_{t+1}} l^f_{t+1}$ is related to the dual purpose of capital in this model, since more capital raises the firms' debt capacity by making repayment more likely. Formally, it consists of the derivative of the loan price with respect to k^f_{t+1}, multiplied by the amount of loans issued. Since this term is always negative, it essentially reduces the cost of acquiring capital.

Loan Pricing. Equation (B.47) reveals that fossil capital requirements affect investment through the loan price schedule and its derivative with respect to k^f_{t+1}. Without restructuring costs but a full loss of output in default, the expected loan payoff simplifies to $\mathcal{R}^f_t = 1 - F(\overline{m}^f_{t+1})$ so that

$$q(\overline{m}^f_{t+1}) = \big((1-\kappa^f_t)\Xi_{t+1} + \overline{\Lambda}_{t,t+1}\big)\big(1 - F(\overline{m}^f_{t+1})\big) , \tag{B.48}$$

where $\overline{\Lambda}_{t,t+1} = \beta(1-G(\mu_{t+1}))$ is the banker sdf. Since the term $G(\mu_{t+1})$ is small and affects the loan pricing condition for all sectors uniformly, we focus on terms related to tilted bank capital requirements and approximate the banker sdf as $\overline{\Lambda}_{t,t+1} = \beta(1-G(\overline{\mu}_{t+1})) \approx \beta$ to obtain (20) in the main text. The derivative of $q(\overline{m}^f_{t+1})$ with respect to the risk choice can be expressed as:

$$\frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} = -\Big((1-\kappa^f_t)\Xi_{t+1} + \beta\Big) f(\overline{m}^f_{t+1}) . \tag{B.49}$$

Capital Requirements and Investment. Quantitatively, it turns out that the expected firm productivity conditional on defaulting $G(\overline{m}^f_{t+1})$ is close to zero and we approximate the first-order condition for investment (B.47) in the following by imposing $G(\overline{m}^f_{t+1}) = 0$. Furthermore, using $\frac{\partial \overline{m}^f_{t+1}}{\partial k^f_{t+1}} = -\frac{\overline{m}^f_{t+1}}{k^f_{t+1}}$ and expanding by $\frac{\widetilde{p}^f_{t+1}}{\widetilde{p}^f_{t+1}}$, we can rearrange for the price of fossil capital:

$$\psi^f_t \approx \Big(\widetilde{\beta} - \frac{\partial q^f_t}{\partial \overline{m}^f_{t+1}} \big(\overline{m}^f_{t+1}\big)^2\Big) \widetilde{p}^f_{t+1} .$$

To link this expression to tilted bank capital requirements, we plug in the derivative of the loan price schedule (B.49) to get

$$\psi^f_t \approx \Big(\widetilde{\beta} + \big((1-\kappa^f_t)\Xi_{t+1} + \beta\big) f(\overline{m}^f_{t+1})(\overline{m}^f_{t+1})^2\Big) \widetilde{p}^f_{t+1} . \tag{B.50}$$

Removing the deposit financing wedge by setting κ^f_t=100% induces a downward shift in the loan price schedule by approximately Ξ_{t+1}. Under the simplifying assumptions, we can approximate the deposit financing wedge as the sum of the liquidity premium and the bank failure rate:

$$
\begin{aligned}
\Xi_{t+1} &= \frac{1}{1+r_t^D} - \frac{1}{1+r_t^{rf}}\left(1 - F(\overline{\mu}_{t+1})\right) , \\
&= \frac{1+r_t^{rf}}{(1+r_t^D)(1+r_t^{rf})} - \frac{1+r_t^D}{(1+r_t^D)(1+r_t^{rf})}\left(1 - F(\overline{\mu}_{t+1})\right) , \\
&= \frac{(1+r_t^{rf}) - (1-F(\overline{\mu}_{t+1}))(1+r_t^D)}{(1+r_t^D)(1+r_t^{rf})} , \\
&= \frac{r_t^{rf} - r_t^D + F(\overline{\mu}_{t+1})(1+r_t^D)}{(1+r_t^D)(1+r_t^{rf})} , \\
&\approx r_t^{rf} - r_t^D + F(\overline{\mu}_{t+1}) ,
\end{aligned}
$$

where we have used that $(1+r_t^D)(1+r_t^{rf}) \approx 1$ and $F(\overline{\mu}_{t+1})r_t^D \approx 0$.

Market Clearing. Lastly, we combine the fossil energy supply condition, equation (B.50), with fossil energy demand. Abstracting from non-energy intermediate goods and labor, there is an simple fossil energy demand function $p_t^f = (1-\nu)\left(\frac{y_t}{z_t^f}\right)^{1/\epsilon}$, which obtains from setting $\alpha = 1$ and $y_t = \widetilde{z}_t = z_t^e$ in the nested CES aggregators of final good producers, see equation (A.42) in Appendix A.5. We impose market clearing and plug the demand for fossil energy together with the definition of $\widetilde{p}_{t+1}^f = p_t^f - \tau_t^e(1-a_t) - \frac{\theta_1}{\theta_2+1}a_t^{\theta_2+1}$ into the first-order condition for fossil capital (B.50) to arrive at

$$
\psi_t^f \approx \widetilde{\beta}\Big\{ \underbrace{\Big((1-\kappa^f)\Xi + \beta\Big) f(\overline{m}^f)(\overline{m}^f)^2}_{\text{Tilted Regulation + Fin. Frictions}} + 1 \Big\} \Big(\underbrace{(1-\nu)\left(\frac{y}{z^f}\right)^{1/\epsilon}}_{\text{Fossil E. Demand}} - \underbrace{\tau^e(1-a) - \frac{\theta_1}{\theta_2+1}a^{\theta_2+1}}_{\text{Tax and Abatement Cost}} \Big) .
$$

In the special case without financial frictions ($\overline{m}_{t+1}^f = 0$) and no investment adjustment cost ($\psi_t^f = 1$), the equilibrium fossil capital share can be written as:

$$
\frac{1}{\widetilde{\beta}} + \tau^e(1-a) + \frac{\theta_1}{\theta_2+1}a^{\theta_2+1} = (1-\nu)\left(\frac{y}{z^f}\right)^{1/\epsilon} .
$$

A higher emission tax increases the LHS, such that the fossil energy share $\frac{z^f}{y}$ on the RHS has to decline.

Capital Requirements and Risk-Taking. We can also use the simplified model to shed light on the sector-specific and aggregate risk-taking effects induced by bank capital requirements and emission taxes. Plugging the loan price schedule and its derivative into the simplified first-order condition for loans, equation (B.46), we obtain

$$\Big((1-\kappa_t^f)\Xi_{t+1}+\beta\Big)\Big(1-F(\overline{m}_{t+1}^f)\Big)-\Big((1-\kappa_t^f)\Xi_{t+1}+\beta\Big)f(\overline{m}_{t+1}^f)\overline{m}_{t+1}^f=\widetilde{\beta}\big(1-F(\overline{m}_{t+1}^f)\big)\ .$$

This condition can be re-arranged to

$$h(\overline{m}_{t+1}^f)\overline{m}_{t+1}^f=\frac{(1-\kappa_t^f)\Xi_{t+1}+\beta-\widetilde{\beta}}{(1-\kappa_t^f)\Xi_{t+1}+\beta} \tag{B.51}$$

where $h(\overline{m}_{t+1}^f)\equiv\frac{f(\overline{m}_{t+1}^f)}{1-F(\overline{m}_{t+1}^f)}$ is the hazard rate of the idiosyncratic productivity shock. By the log-normal assumption on idiosyncratic productivity, the hazard rate is monotonically increasing in $\overline{m}_{t+1}^f$. The RHS of equation (B.51) is also increasing in the term $(1-\kappa_t^f)\Xi_{t+1}$ associated with the deposit financing wedge and the capital requirement. Taken together, this implies that fossil energy firms reduce their risk-taking if banks need to fossil loans by equity ($\kappa^f=100\%$). It also follows that risk-taking increases if the deposit-financing wedge Ξ_t widens.

C Data Sources and Calibration

Central Bank Speeches. Figure 1 is based on the database provided by Campiglio et al. (2025), which is available online under `https://cbspeeches.com/`. We extract the number of speeches by central bank governors, board members and senior officials that mention `Climate Change` and `Capital Requirements` in the same speech on obtain 205 matches. Of those speeches, 26 are by Bank of England, 19 by the European Central Bank, 18 by the Banque de France, 14 by the Banca d'Italia, 20 by the Banco d'Espana, 31 by the Deutsche Bundesbank, 28 by other Eurosystem central banks, 7 by the Federal Reserve and 17 by officials across various Emerging Economies. This fairly even distribution across regions suggests a broad interest in the topic in particular after 2019.

Notably, the topic has received very little attention across all regions before Mark Carney's speech "Breaking the tragedy of the horizon - climate change and financial stability" in 2015. Interest intensified in 2018, for example by a speech titled "Climate change and central banking" by ECB board member Yves Mersch in 2018 and by a speech with the same title by Bundesbank governor Jens Weidmann in 2019. In recent years, speeches address the interplay between central banking, financial markets and climate change explicitly, for example the speech "Building an appropriate climate stress testing framework for capital markets" by Banque de France deputy governor Denis Beau in 2021 and the speech "Green energy and green finance for a sustainable Europe" by Bundesbank board member Sabine Mauderer in 2022. This time pattern is consistent with the recent surge in climate related financial policies as documented by Rischen (2025).

Data Sources. Table C.1 summarizes the data sources used in the calibration. Whenever data are available, we focus on the period 1984-2013 and obtain data at annual frequency. Leverage is available at quarterly frequency until 2012 from the replication data provided by Gomes et al. (2016). To obtain an annual time series, we simply average corporate leverage ratios within each year. Corporate default rates are also provided by Gomes et al. (2016) and we use the default frequency of all rated firms to construct our data moments. Recovery rates are provided by the World Bank, following the methodology outlined in Djankov et al. (2008). We take the simple time series average for the US, which is however only available after 2013. Data for several OECD countries (Australia, Austria, Canada, Germany, Denmark, Finland, France, Netherlands, Norway, South Korea, Sweden, United Kingdom) is available since 2003 and average across countries and years also yields a value of 82%.

Variable	Source	Series/Ticker
Bank Failure Rate	FDIC	Total Commercial Banks Failures: Assisted Merger Failures: Paid Off
Corporate Default	Gomes et al. (2016)	All Rated Firms
Corporate Leverage	Gomes et al. (2016)	Net Debt over Net Assets
Corporate Recovery Rate	World Bank	Recovery rate (cents on the dollar)
Real GDP	St Louis Fed	GDPCA
Nominal GDP	St Louis Fed	NGDPXDCUSA
Nominal Deposits	St Louis Fed	SAVINGSSL
Inflation	St Louis Fed	FPCPITOTLZGUSA
Federal Funds Rate	St Louis Fed	RIFSPFFNA

Table C.1: Data: Sources and Ticker.

In Figure C.1, we plot key financial market variables against the cyclical component of real GDP. Corporate leverage ratios, default rates, and bank failure rates show a strong comovement with each other and tend to peak around recessions. In particular, corporate sector leverage appears to lead the corporate default rate, the bank failure rate, and economic downturns. Schularick and Taylor (2012) provide extensive evidence along these lines. In the model, this is reflected by firms' risk-taking decision that induces corporate leverage to increase prior to the default and bank failure rates. Consequently, these variables (leverage in particular) exhibit a counter-cyclical behavior in our model as well, see Table 1.

Comparison to Euro Area Data. In our calibration, we target US variables to ensure that we match the model to an internally consistent set of data moments. However, it is not the purpose of our paper to provide policy recommendations tailored to the US but rather to demonstrate the macroeconomic effects of tilted bank capital regulation in

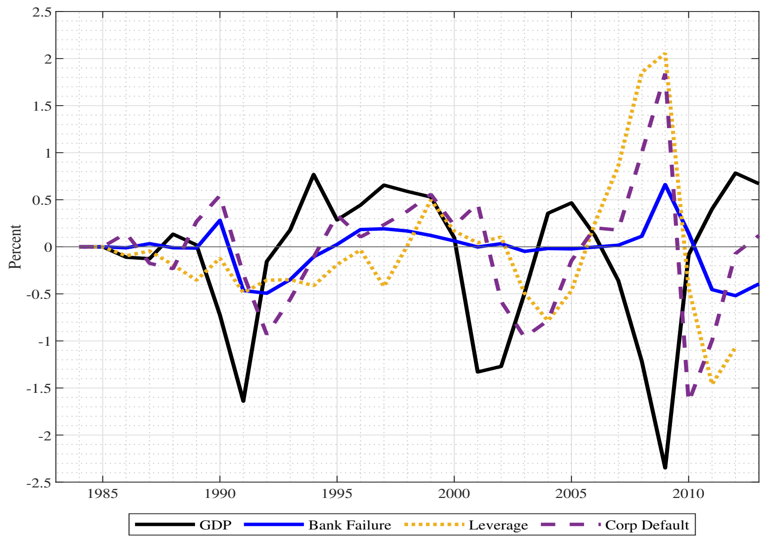

Figure C.1: Bank Failure and Corporate Default Rates: Annual real GDP is logged and detrended using a HP-filter with smoothing parameter 6.25.

a reasonably calibrated model. To corroborate broader applicability of our quantitative analysis, it is helpful to compare our data moments to the moments used by Mendicino et al. (2024), who calibrate a similar model with bank failure and corporate default to euro area data.

Moment	US	Euro Area
Corporate default rate $ave(F(\overline{m}))$	1.6	2.6
Bank failure rate $ave(F(\overline{\mu}))$	0.6	0.66
Relative vol. investment $\sigma(i)/\sigma(y)$	1.50	2.01
Relative vol. consumption $\sigma(c)/\sigma(y)$	0.88	0.81
Relative vol. corporate default $\sigma(F(\overline{m}))/\sigma(y)$	0.50	1.59
Relative vol. bank failure $\sigma(F(\overline{\mu}))/\sigma(y)$	0.33	1.21
Correlation Firm default-Bank failure	0.56	0.64

Table C.2: Comparison between US and Euro Area Data.

As Table C.2 shows, typical second moments used in the business cycle literature, such as the excess volatility of consumption and investment over GDP are quite similar. Concerning financial markets, the bank failure rate is almost identical, while the corporate default rate is around one percentage point larger in the euro area. While the euro area also exhibits a larger excess volatility of default risk and bank failure, the correlation between bank failure and firm default is very similar at 64% compared to 56%. This is a key untargeted moment for our analysis as it connects default risk in the non-financial sector, which is induced by emission taxes in our setting, to bank failures, which is crucial

Parameter	**Value**	**Source**
Households		
Household discount factor β	0.99	Standard
Consumption CRRA γ_C	2	Standard
Liquidity curvature γ_D	1.5	Target: Excess vol. deposit/GDP
Liquidity weight ω_D	0.0074	Target: Liquidity premium
Labor supply curvature γ_N	1	Frisch elasticity of one
Labor supply weight ω_N	20	$n^{SS} = 0.33$
Technology		
Cobb-Douglas coefficient α	1/3	Standard
Capital depreciation rate δ_K	0.08	Standard
Inv. adj. parameter Ψ_I	2	Target: Excess vol. investment
Price adjustment cost Ψ_P	4.5	5 quarters price duration
Final good CES ϕ	6	20% Price Markup
Sticky Wage Share Ψ_N	0.63	Del Negro et al. (2015)
Labor CES $\widetilde{\phi}$	3	Smets and Wouters (2003)
Energy weight $\widetilde{\nu}$	0.0015	Energy share 10%
Energy/non-energy CES $\widetilde{\epsilon}$	0.2	Standard
Clean weight ν	0.3865	Fossil energy share 80%
Fossil/clean energy CES ϵ	3	Papageorgiou et al. (2017)
Abatement cost parameter θ_1	0.05	Full Abatement Tax 150$/tCO2
Abatement cost parameter θ_2	1.6	Heutel (2012)
Emission damage parameter Ψ_E	0.1	Damage/GDP ratio
Financial Markets and Policy		
Firm-owner discount factor $\widetilde{\beta}$	0.988	Target: Firm leverage
St. dev. firm risk ς_m	0.2	Target: Firm default rate
Loan maturity parameter χ	0.2	5-year average maturity
St. dev. bank risk ς_μ	0.0315	Target: Bank failure rate
Restructuring costs φ	0.12	Target: Recovery rate
Bailout loss ζ	0.0115	Optimality of κ^{sym}
Capital requirement κ^{sym}	0.08	Basel II
Taylor rule coefficient φ_π	2	Target: Excess vol. inflation
Shocks		
MP shock st. dev. σ_R	0.015	Target: Excess vol. nominal rate
Persistence TFP ρ_A	0.8	Standard
TFP shock st. dev. σ_A	0.004	Standard
Persistence Tax ρ_T	0.8	Standard
Tax shock st. dev. σ_T	0.0025	5$/tCO2 st. dev.

Table C.3: Baseline Calibration.

for bank capital regulation. Table C.3 summarizes all parameters from our baseline calibration.

Mapping Model Emission Taxes into the Data. The model-implied emission tax is converted into \$/tCO2 by converting model units of output and emissions into their real world counterparts. As we abstract from rest-of-the-world emissions, we map $y^{model} = 0.441$ in the steady state without emission taxes into global GDP ($y^{world} = 105.00$ trillion USD in 2022). In a similar fashion, we convert model emissions ($e^{model} = 0.318$ in the baseline calibration) into global emissions ($e^{world} = 37.5$ GtCO2 in 2022). From the expression $p^e \cdot \frac{e^{\text{world}}}{y^{\text{world}}} = \tau^e \cdot \frac{e^{\text{model}}}{y^{\text{model}}}$, the model-implied emission price in \$/tCO2 is given by $p^e = \frac{y^{\text{world}}/y^{\text{model}}}{e^{\text{world}}/e^{\text{model}}} \tau^e = 2021 \cdot \tau^e$.

To compute the global coverage-weighted emission tax, we obtain data from the World Bank's carbon pricing dashboard, available at this link. This dataset provides annual carbon taxes and prices of emission certificates as well as the share of global emissions covered by each tax or emission pricing scheme. This yields a value of 5.40\$/tCO2.

D Additional Numerical Results

In this section, we show the effects of tilted bank capital requirements (D.1), emission taxes (D.2) and (optimal) symmetric bank capital requirements (D.3) on additional welfare-relevant variables. Appendix D.4 provides further details on the role of tilted bank capital requirements for the short run effects of the net zero transition.

D.1 Fossil and Sustainability Linked Capital Requirements

In Section 4 in the main text, we have shown how fossil penalizing capital requirements and sustainability linked capital requirements affect key welfare-relevant variables, such as the reduction of socially harmful emissions, the bank failure rate and the liquidity premium on deposits. Figure D.1 displays additional aggregate and sectoral effects of tilted bank capital requirements. As in Figure 2, we vary the simple fossil capital requirement (dashed red line) and the penalty capital requirement κ_t^{pen} on the x-axis. We set $\kappa_t^{reward} = \kappa_t^c = 8\%$ (dotted green line), i.e. firms initially using the fossil technology are treated like clean energy firms if they choose to abate all their emissions ($a_t = 1$).

The top row demonstrates that consumption and GDP increase for the simple fossil requirement, which is largely because this policy eliminates the possibility of bank failure. The SLC induces a stronger increase, because it also reduces emission damages more effectively. In the second row, we illustrate the sectoral re-allocation away from fossil towards clean and non-energy firms. While there is a reduction in capital for initially fossil firms by around one percent, clean firms accumulate around 5% more capital, while capital of non-energy firms increases by around 0.4%.

The effects are generally stronger for loans than for investment. This directly follows from the first-order condition for leverage (14), which implies that a higher fossil capital requirement induces fossil energy firms to switch to more equity financing. The reason is that higher capital requirements worsen firms' loan financing conditions. Hence, fossil loans decline much more than fossil capital, which is accompanied by a decline in the fossil default rate by more than one percentage point, see the fourth row of Figure D.1.

This side effect of fossil capital requirements on the fossil default rate is shown formally in equation (B.51), Appendix B.

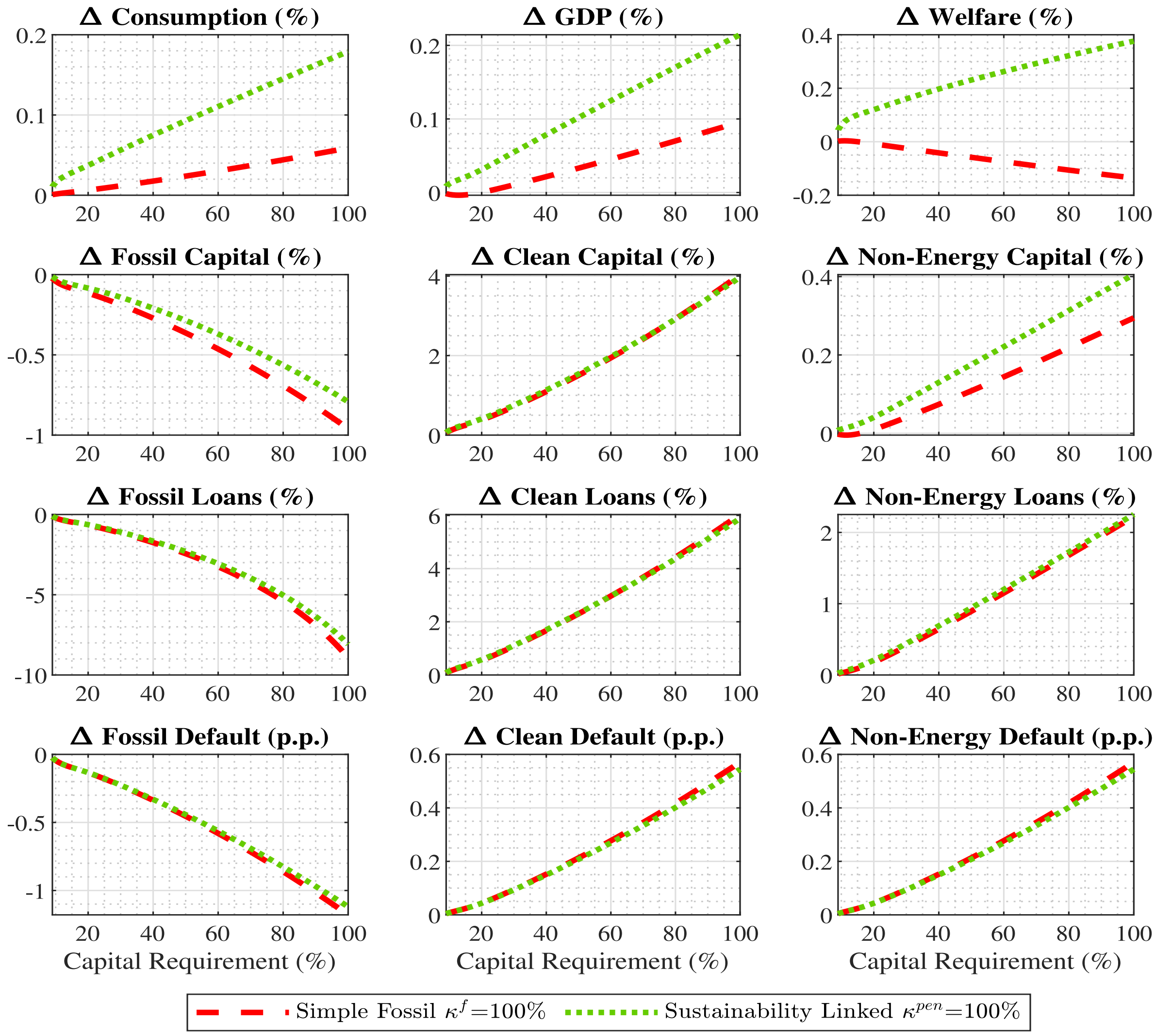

Figure D.1: Sectoral Effects of Tilted Capital Requirements. This figure displays additional welfare-relevant aggregates in the top row. The bank failure rate is expressed in percent, consumption and GDP are expressed relative to the baseline without emission taxes. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (in percentage points).

As we have demonstrated in Figure 2, higher fossil capital requirements are associated with a contraction in deposits and hence a larger deposit financing wedge Ξ_{t+1}. As in the model of Begenau (2020), this reduces loan price schedules in *all* sectors. On one hand, this counteracts the reduction in default risk by the fossil sector. On the other hand, it induces the default rate in other sectors to increase by more than half a percentage point, which is sizable given that its initial level is 1.7%. We again refer to Appendix B for a formal analysis of this effect. A rise in loan price schedules also implies that investment in the clean and non-energy sectors rises. This is in sharp contrast to the sectoral spillovers of emission taxes, which would depress non-energy investment, as energy and non-energy intermediate goods are complements, see also Figure D.2. Hence, the uptake in investment is not primarily a desirable sectoral re-allocation but comes at the cost of elevated risk-taking in the non-financial sector.

D.2 Emission Taxes and Climate Policy Uncertainty

In this section, we show in detail how welfare-relevant variables are affected by emission taxes and emission tax shocks.

Emission Taxes. First, we consider the effects of emission taxes in the baseline model (solid blue line) and in an extension with policy uncertainty (dashed magenta line) in Figure D.2. The capital requirement is symmetric across sectors and fixed at 8%. The top row shows the three key welfare-relevant variables. A 100\$/tCO2 tax reduces emissions to zero. The marginal effect of taxes on emissions is larger if the tax level is low, i.e. firms first pick the "low-hanging fruits" when it comes to de-carbonizing the energy sector, which follows from the convex specification of abatement costs.

Importantly, climate policy does not affect the bank failure rate when taxes are constant as the regulator essentially chooses the bank failure rate by setting capital requirements. As the top right panel of Figure D.2 illustrates, the liquidity premium widens by around 3bps, which directly follows from a contraction in aggregate loan demand. The second row of Figure D.2 shows that consumption and GDP increase if climate policy is more stringent. This is primarily due to reduction in damages associated with the emission externality and results in a large welfare gain of achieving net zero emissions.

The third and fourth row decomposes the aggregate loan demand contraction into its sectoral origins. As for tilted bank capital requirements, stringent climate policy lower fossil capital and loans, while clean capital and loans increase. Generally, non-energy capital is lower under full abatement due to the low substitution elasticity between energy and non-energy goods. However, there is an equilibrium effect operating against the reduction of non-energy capital and against the aggregate loan demand contraction. By increasing deposit scarcity, the deposit financing advantage widens, which lowers bank funding costs and increases loan price schedules, similar to the adverse side effects of tilted bank capital requirements.

The associated outward shift in loan price schedules induces investment, loan demand and risk-taking to rise across sectors. This essentially reinforces the positive investment

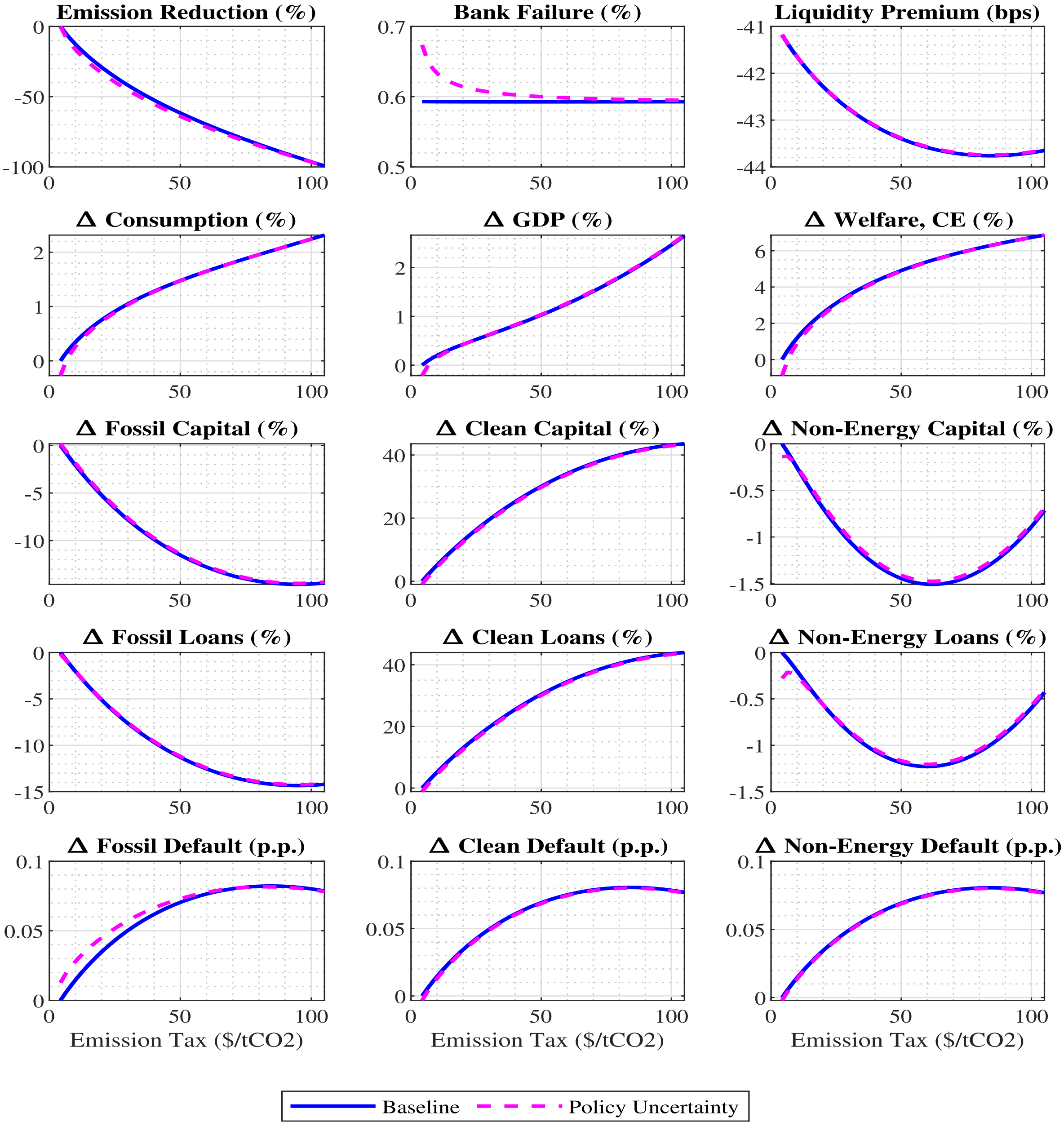

Figure D.2: Long Run Effects of Emission Taxes. This figure displays additional welfare-relevant aggregates in row one and two. The bank failure rate is expressed in percentage points, consumption and GDP are expressed relative to the baseline without emission taxes. The third, fourth and fifth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (annualized in percent). The emission tax on the x-axis is expressed in $/tCO2.

and loan demand effects in the clean sector overturns the negative investment and loan demand effects in the non-energy sector, and it translates into a 0.1 percentage point increase in the default rate, see the fifth row of Figure D.2. Note that there are no heterogeneous effects on risk taking, which we observed for tilted bank capital requirements on sectoral default rates. Despite the reduction in liquidity supply the welfare gain of emission taxes is increasing in τ_t^e and quantitatively large.

Climate Policy Uncertainty. The dotted magenta line in Figure D.2 represents the same changes in long run emission taxes subject to climate policy uncertainty in the form of tax shocks. The upper row of Figure D.2 shows that climate policy uncertainty further reduces investment in the fossil sector and hence aggregate emissions decline more for every emission tax level. These effects are consistent with the literature, see for example Fried et al. (2022).

Consistent with Hambel and van der Ploeg (2025), fossil energy firms exhibit a larger default rate in the presence of emission tax shocks, while the default rate of clean firms declines slightly. Crucially, the realized default rates $F(\overline{m}_t^f)$ and $F(\overline{m}_t^c)$ are not linear in the respective default thresholds. Hence the fossil default rate increases by more than the decline in the clean default rate. Hence, the aggregate default rate and the bank failure rate rise. Taken together, climate policy uncertainty reduces welfare. For low emission tax levels, climate policy uncertainty even reduces welfare relative to the baseline model without emission taxes. Notably, this effect vanishes if the emission tax level is higher, as fossil firms choose a higher abatement level and are hence less susceptible to emission tax shocks.

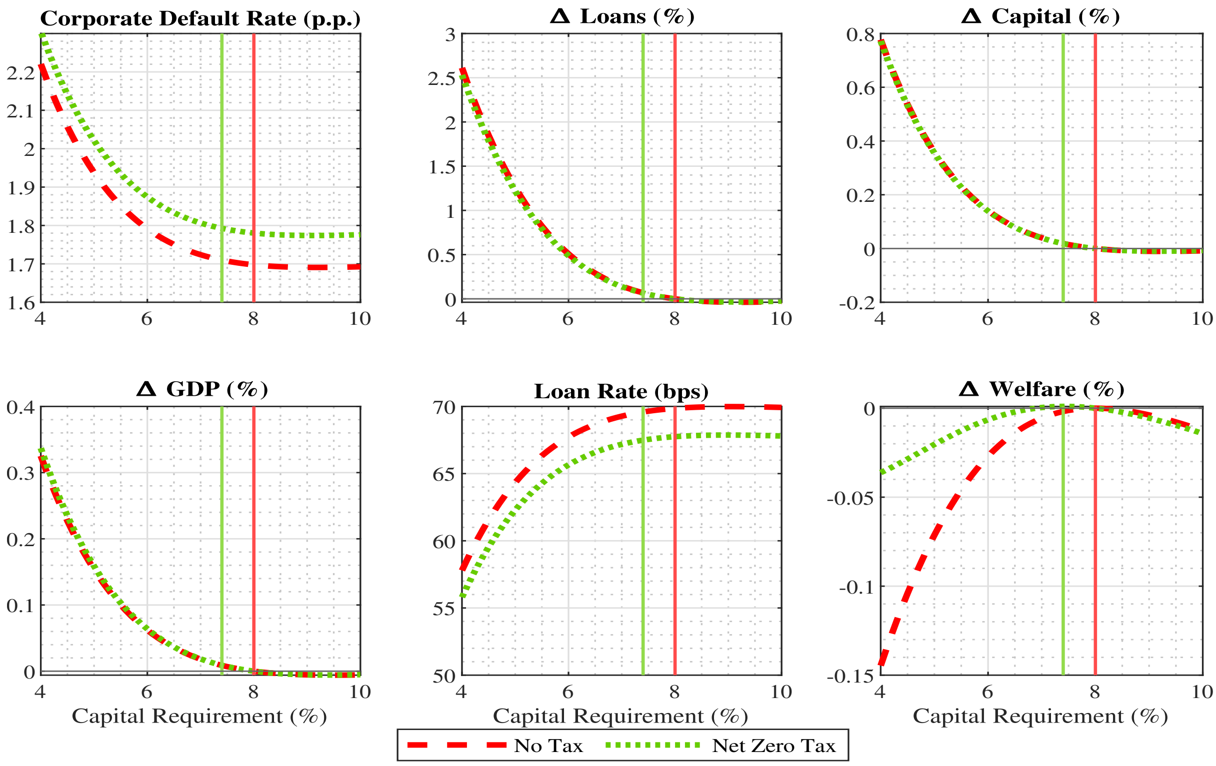

Figure D.3: Symmetric Capital Requirements, Emission Tax Level. The corporate default rate is expressed in percentage points, loans, capital, GDP and emissions are expressed relative to the baseline without emission taxes in percent. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (in percent). Welfare is defined in equation (A.31) and expressed in consumption equivalent relative to the baseline without climate policies.

D.3 Optimal Capital Requirements

Figure 4 in the main text shows how symmetric bank capital requirements affect the bank failure rate and the liquidity premium on deposits for different emission tax levels and in the case of climate policy uncertainty. The bank failure rate and the liquidity premium are pivotal for the optimal symmetric bank capital requirement. In this section, we display the effect of κ^{sym} on investment and risk-taking in the non-financial sector, which also have first-order effects on welfare and are hence relevant for optimal capital requirements.

The Transmission of Bank Capital Requirements. We first discuss the transmission of symmetric capital requirements in the absence of climate policy, represented by the dashed red line in Figure D.3. The top row shows that a reduction in κ^{sym} increases aggregate investment, loans, and the corporate default rate. This directly follows from the outward shift of the loan price schedule, where the expansionary effect is a consequence of higher weight $(1-\kappa_t^{sym})$ on the deposit-financing wedge Ξ_{t+1} in banks' loan pricing condition (5). Firms increase their risk choice $\overline{m}_{t+1}$ in response to this outward shift in the loan price schedule, see Appendix B for a formal derivation.

As we have discussed in the main text, Section 5, favorable loan price schedules also increases the incentives to invest, since more investment enhances firms' debt capacity and makes borrowing more attractive. As the bottom row of Figure D.3 illustrates, the increase in investment is related to an increase in GDP. Notably, the associated expansion of investment and GDP comes at the expense of substantially larger bank bailout and corporate default costs and is not optimal. Taking all these effects into account, the bottom right panel shows that welfare is concave and exhibits an unique interior maximum. It is attained at 8% by the design of the bailout cost parameter, which is calibrated to render an 8% capital requirement optimal.

Figure D.3 also allows us to assess the external validity of our model. As no country has yet implemented tilted bank capital requirements, we compare the implied effects of changing symmetric bank capital requirements to empirical estimates from the literature. To ensure a fair comparison it has to be considered that i) loan rates, loan volumes and investment depend non-linearly on the bank capital requirement in the model and, ii) our model always imposes a 100% risk weight on bank assets. In practice, banks hold less regulatory equity due to the possibility of risk weights that are smaller than 100%. Against this background, it appears reasonable to compare the effects of changing κ^{sym} from our baseline value of 8% to the effects of changing it from a lower value, say 5% or 6%.

In the bottom middle panel of Figure D.3, we observe that lowering κ^{sym} from 5% to 4% induces a decline of the loan rate by around 8bps, which is consistent with the empirical effects of a one percentage point decline in Glancy and Kurtzman (2021). Lowering κ^{sym} from 5% to 4% would induce loan volumes to increase by around one percent, consistent with empirical findings by De Marco et al. (2021), see the top middle panel. In the bottom left panel of Figure D.3, we observe that the increase in investment translates into a higher level of GDP. To assess the quantitative plausibility of the effect size, we

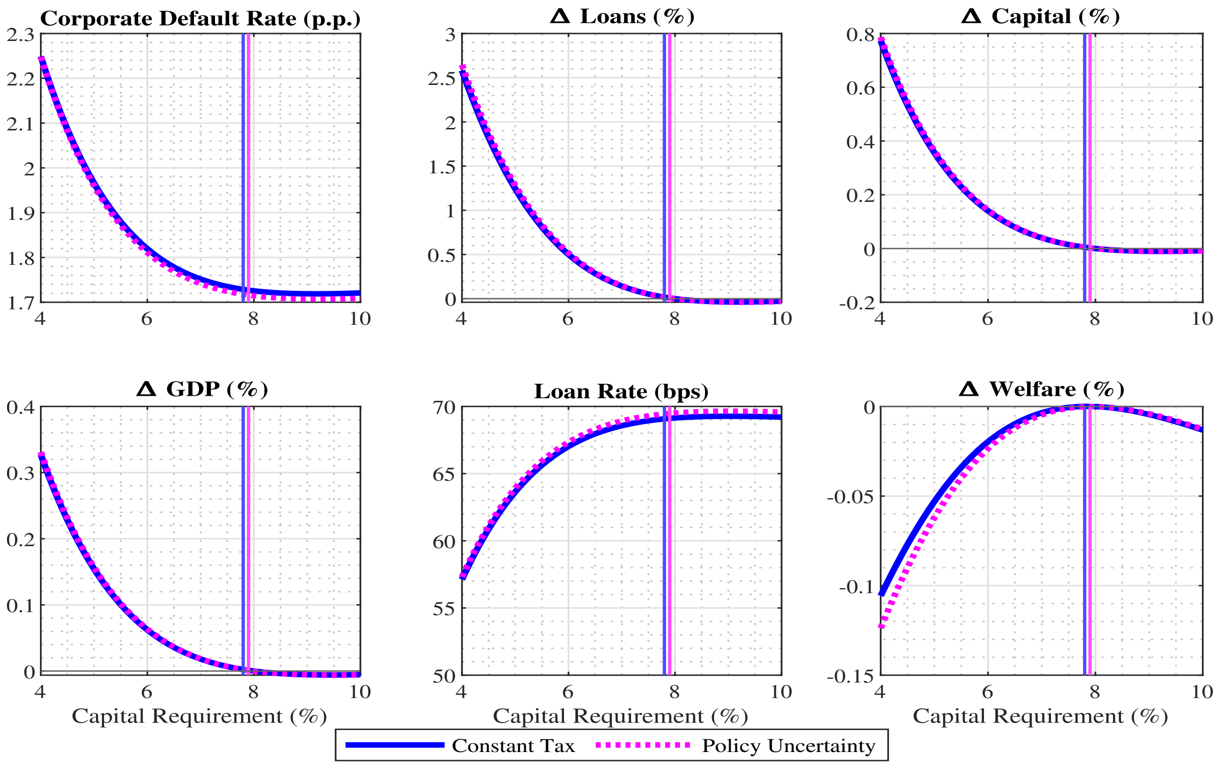

Figure D.4: Symmetric Capital Requirements, Emission Tax Shocks. The corporate default rate is expressed in percentage points, loans, capital, GDP and emissions are expressed relative to the baseline without emission taxes in percent. The second, third and fourth row compare the sectoral and aggregate effects of emission taxes in terms of investment, loans and default rates (in percent). Welfare is defined in equation (A.31) and expressed in consumption equivalent relative to the baseline without climate policies.

draw from a meta analysis by Fidrmuc and Lind (2020), who find that a one percentage decline in the capital requirement induces a 0.09% increase in GDP. In the model, lowering κ^{sym} from 8% to 7% raises GDP by merely 0.03%, while lowering it from 5% to 4% increases GDP by around 0.17%, which appears reasonable in comparison to the data. Take together, the real effects of bank capital requirements in our model are in line with the empirical literature.

Emission Tax Level. How does the climate policy regime affect the trade-off between bank failure and the deposit channel of bank regulation? As we have shown in the main text (Figure 4), emission taxes have no effect on bank failure rates and no *direct* effect on the corporate default rate. However, they have an adverse side effect on liquidity provision to households which is associated with a modest increase in the aggregate default rate. Consequently, the optimal bank capital requirement declines by 0.6 percentage points from 8% to 7.4%.

Climate Policy Uncertainty. In Figure D.4, we explore the role of emission tax shocks in greater detail. The solid blue line refers to a constant tax of 5$/tCO2, while the dashed magenta line refers to the same tax level with emission tax shocks. We observe a very similar shape across all variables, suggesting that the presence of climate policy uncertainty does not affect the transmission of bank capital requirements. Furthermore, the loan rate, investment and GDP are very similar, consistent with the very small effects on the liquidity premium presented in Figure 4 in the main text. As the upper panel shows, the corporate default rate is consistently a few basis points higher if emission taxes are subject to uncertainty, which then gives rise to the larger bank failure rate and renders a higher bank capital requirement optimal. Interestingly, this is only relevant as long as the fossil sector is highly susceptible to emission tax shocks. The negative effect of climate policy uncertainty on the bank failure rate vanishes for higher tax levels, since firms have adopted emission-free technologies more widely in this case.

D.4 Tilted Bank Regulation and the Net Zero Transition

In this section, we display additional macro-financial variables along the net zero transition, complementing Figure 5. Solid blue lines indicate the baseline transition where emission taxes are the only climate policy instrument. The top row of Figure D.5 shows that the share of abated emissions increases from its initial level of 33% in 2025 to around 66% in 2035 and reaches net zero in 2050 by the construction of the tax path. GDP increases along the transition as higher emission taxes reduce economic damages associated with emissions, see the top right panel. The dotted green line reveals that the SLC has a positive effect on emissions and GDP by inducing higher emission abatement. In addition, GDP would be even higher as the SLC is also associated with a drastic decline in bailout costs, which follows from much fewer bank failures.

The second row of Figure D.5 displays the sectoral shifts from fossil towards clean capital. As different energy sources are easily substitutable, clean investment rises by around 12%, while fossil investment declines by 7% over the next decade. Importantly, energy goods are not *perfectly* substitutable such that total energy investment and production decline. Since energy and non-energy intermediate goods are complements, non-energy investment is depressed as well. The capital re-allocation from fossil to clean capital is slightly more pronounced if the SLC is in place.

The third row of Figure D.5 illustrates that loans in each sector closely follow sector-specific capital. As a consequence, aggregate loan demand contracts, which translates into an elevated liquidity premium associated with less deposit supply. This reduces banks' funding cost and increases the deposit financing wedge Ξ_t in the loan pricing condition (5) across all sectors.

This general equilibrium effect slightly counteracts the aggregate loan contraction but comes at the cost of elevated risk-taking. Indeed, loans decline by less than capital, i.e. the leverage ratio and default risk increases in all sectors over the medium run. This is illustrated in the last row of Figure D.5. In fact, this even increases loans in the non-energy sector, at least temporarily. However, the overall productivity losses associated with abatement spending eventually dominate such that non-energy loans decline slightly

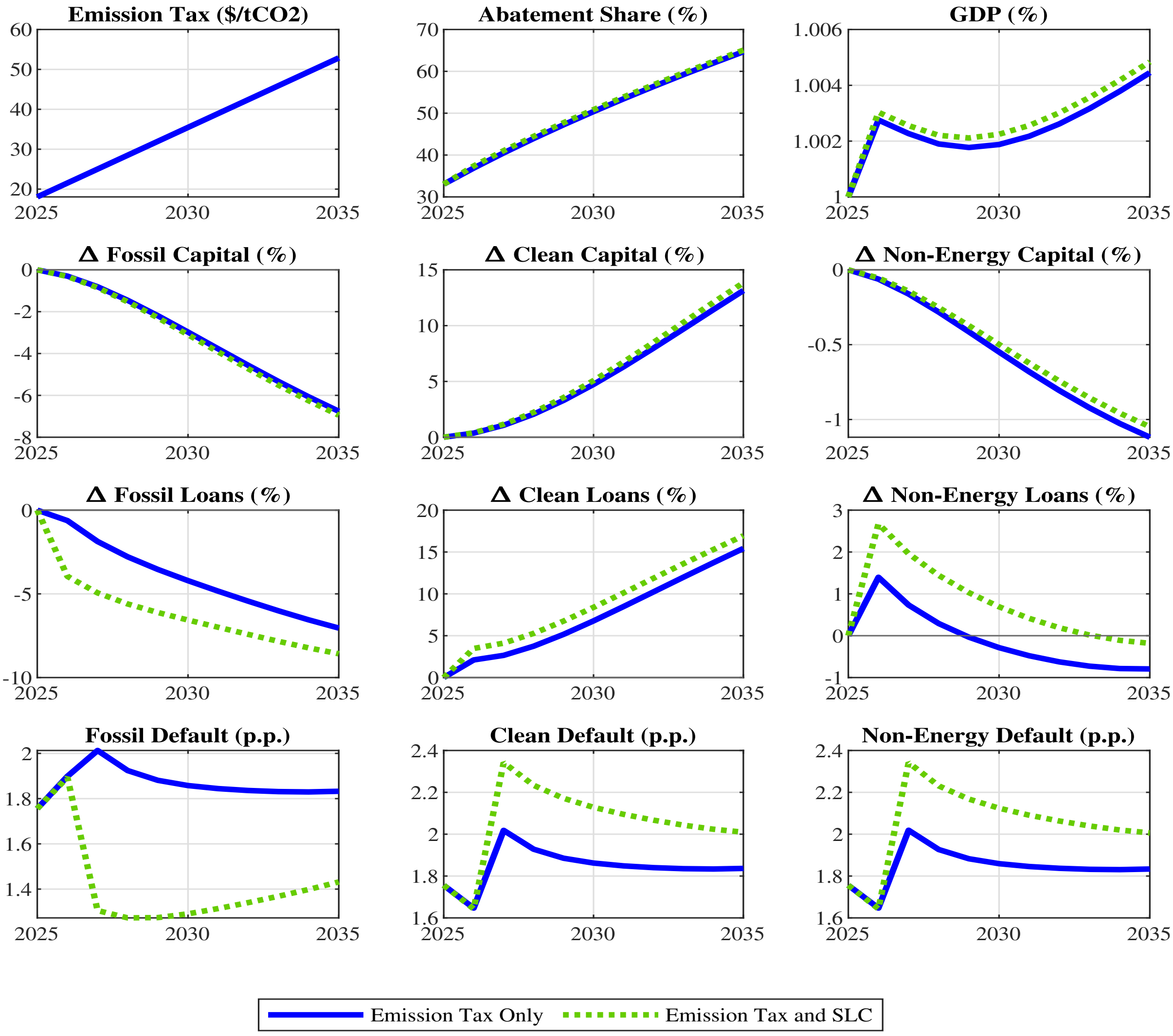

Figure D.5: Transition Path: Additional Variables. Linear emission tax path, starting from 5$/tCO2 in 2025 to 100$/tCO2 in 2050. Adding sustainability linked capital requirements to the policy mix is reflected by the dotted green line. The changes in sector-specific capital and loans are computed relative to the initial year 2025.

in the medium term.

The medium run rise in default risk across all three sectors is a direct consequence of the increased loan take-up. In addition, sector-specific default risk is affected directly by the emission tax. The fossil default threshold $\overline{m}_t^f$ increases on impact so that more fossil firms default in the first period of the transition. By contrast, clean and non-energy firms benefit as demand for their output increases, which results in a temporary decline of $\overline{m}_t^c$ and $\overline{m}_t^f$. This heterogeneous short-run effect is responsible for the non-monotonic effects of the transition on clean and non-energy default rates and the hump in fossil default risk shown in the bottom row of Figure D.5.

E Robustness

In this section, we provide additional details to the robustness checks in Section 7.

E.1 Households and Climate Block

The first panel of Table 3 displays robustness checks with respect to climate and household parameters and the specification of the abatement technology. Here, we discuss the implications of these changes in greater detail and derive the first-order condition for abatement if it is modelled as an investment choice.

Higher Abatement Cost. We first consider the case of higher emission abatement cost for firms initially using the fossil technology. Specifically, we set θ_1=0.1, representing a considerable increase from the baseline value of θ_1=0.05. Under this parameter, 1.9% of GDP are spent on abatement to induce net zero emissions compared to 1.2% in the baseline calibration. Panel B of Table E.1 shows that the transmission of simple fossil capital requirements does not depend on θ_1, since the wedge in the expected payoff from fossil investment only depends on parameters related to financial frictions. For the same reason, the SLC is much less effective than in the baseline calibration. Its benefits only depend on financial frictions, while the cost are much larger in this case. Interestingly, abatement costs are so large that reducing taxes to their net zero level (200\$/tCO2) is no longer optimal. The welfare gain amounts to one percent compared to seven percent in the baseline, while the welfare gain of small emission taxes are larger. Nevertheless, the effects of climate policy (uncertainty) on liquidity provision and bank failure rates are remarkably similar to the baseline case.

Abatement Investment. In the baseline model, the choice to abate emissions is made period by period, similar to a labor demand decision. While this appears to be a reasonable assumption, which is also in line with the E-DSGE literature, the decision to mitigate emissions can also be interpreted as an investment decision that is made before emission taxes realize. Re-interpreting abatement as an investment decision also interacts

	(1) No policy	(2) κ^f=100%	(3) SLC	(4) Net zero tax	(5) 5$ tax	(6) Tax shocks
Panel A: Baseline Model						
Δ Emissions	-	-0.96%	-4.59%	-100%	-18.02%	-12.01%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-59bps	-58bps	-43bps	-41bps	-41bps
Welfare Gain, CE	-	-0.13%	+0.38%	+6.93%	+1.97%	+1.26%
Panel B: High Abatement Cost						
Δ Emissions	-	-0.96%	-3.31%	-100%	-12.10%	-8.75%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.61%
Liquidity Premium	-4bps	-59bps	-58bps	-44bps	-41bps	-41bps
Welfare Gain, CE	-	-0.13%	+0.20%	+1.12%	+1.28%	+0.81%
Panel C: Abatement Investment						
Δ Emissions	-	-0.97%	-4.53%	-100%	-17.10%	-11.91%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.61%
Liquidity Premium	-41bps	-59bps	-58bps	-41bps	-41bps	-41bps
Welfare Gain, CE	-	-0.14%	+0.36%	+6.95%	+1.98%	+1.23%
Panel D: High Climate Damages						
Δ Emissions	-	-0.92%	-4.41%	-100%	-16.73%	-11.70%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.74%
Liquidity Premium	-41bps	-59bps	-57bps	-43bps	-40bps	-40bps
Welfare Gain, CE	-	-0.02%	+0.96%	+21.97%	+4.28%	+2.79%
Panel E: High Deposit Valuation						
Δ Emissions	-	-1.50%	-6.16%	-100%	-17.15%	-12.02%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.41%
Liquidity Premium	-61bps	-89bps	-87bps	-65bps	-62ps	-62bps
Welfare Gain, CE	-	-0.21%	+0.46%	+7.08%	+2.02%	+1.27%

Table E.1: Robustness: Climate and Household Parameters. We compare the case of no policy (zero emission taxes and symmetric capital requirements in column 1) to different climate policies. Column (2) refers to a simple fossil penalizing capital requirement (κ^f=100%), column (3) to the sustainability linked capital requirement with κ^{pen}=100% and κ^{reward}=8%, column (4) to the emission tax consistent with net zero, column (5) to a constant 5$/tCO2 tax and column (6) to an average 5$/tCO2 tax subject to emission tax shocks. The emission change and welfare are computed relative to the baseline. Welfare is defined in equation (A.31), computed relative to the baseline without climate policies and expressed in consumption equivalents.

non-trivially with financial frictions in our model.

While the relationship between emissions and output is still given by $e_t = (1 - a_t)z_t^f$, firms choose next period's abatement a_{t+1} in period t. The fossil energy price net of emission tax spending is given by $\widetilde{p}_t^f = p_t^f - \tau_t^e(1 - a_t)$ and the partial derivative of the default threshold $\overline{m}^f_{t+1} = \frac{\chi l^f_{t+1}}{\pi_{t+1}\widetilde{p}^f_{t+1}k^f_{t+1}}$ with respect to a_{t+1} reads

$$\frac{\partial \overline{m}^f_{t+1}}{\partial a_{t+1}} = -\overline{m}^f_{t+1}\frac{\tau^e_{t+1}}{\widetilde{p}^f_{t+1}} \,. \tag{E.52}$$

We assume that $\frac{\theta_1}{1+\theta_2}(a_{t+1})^{1+\theta_2}$ units of the final good can be transformed into a_{t+1} units of the abatement good. Dividends in period t are now given by

$$\begin{aligned} div^f_t = &\int_{\overline{m}^f_t}^{\infty} \widetilde{p}^f_t m_t k^f_t - \chi\frac{l^f_t}{\pi_t} dF(m_t) + q(\overline{m}^f_{t+1})\Big(l^f_{t+1} - (1-\chi)\frac{l^f_t}{\pi_t}\Big) \\ &\qquad - \frac{\theta_1}{1+\theta_2}(a_{t+1})^{1+\theta_2}k^f_{t+1} - \psi^f_t(1+\tau^f)i^f_t \,, \end{aligned}$$

and the maximization problem now reads

$$\begin{aligned} \max_{a_{t+1},k^f_{t+1},l^f_{t+1},\overline{m}^f_{t+1}} \; & -\psi^f_t(1+\tau^f)k^f_{t+1} - \frac{\theta_1}{1+\theta_2}(a_{t+1})^{1+\theta_2}k^f_{t+1} + q(\overline{m}^f_{t+1})\Big(l^f_{t+1} - (1-\chi)\frac{l^f_t}{\pi_t}\Big) \\ & + \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\int_{\overline{m}^f_{t+1}}^{\infty} \widetilde{p}^f_{t+1} m k^f_{t+1} - \chi\frac{l^f_{t+1}}{\pi_{t+1}} dF(m) - \frac{\theta_1}{1+\theta_2}(a_{t+2})^{1+\theta_2}k^f_{t+2} \\ & \qquad + \psi^f_{t+1}(1+\tau^f)(1-\delta_K)k^f_{t+1} + q(\overline{m}^f_{t+2})\Big(l^f_{t+2} - (1-\chi)\frac{l^f_{t+1}}{\pi_{t+1}}\Big)\Big) \,. \end{aligned}$$

While the first-order conditions for debt and the risk choice remain unchanged, the first-order conditions for capital and abatement become

$$\begin{aligned} &\psi^f_t(1+\tau^f) + \frac{\theta_1}{1+\theta_2}(a_{t+1})^{1+\theta_2} - \lambda^f_t\frac{\overline{m}^f_{t+1}}{k^f_{t+1}} \\ &\qquad = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\psi^f_{t+1}(1-\delta_K)(1+\tau^f) + \widetilde{p}^f_{t+1}\Big(1 - G(\overline{m}^f_{t+1})\Big)\Big) \,. \end{aligned} \tag{E.53}$$

$$\theta_1(a_{t+1})^{\theta_2}k^f_{t+1} - \lambda^f_t\overline{m}^f_{t+1}\frac{\tau^e_{t+1}}{\widetilde{p}^f_{t+1}} = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\big(1 - G(\overline{m}^f_{t+1})\big)\tau^e_{t+1}k^f_{t+1} \,. \tag{E.54}$$

The possibility of default affects the first-order condition for abatement (E.54) through to opposing channels. Optimal abatement is now taking into account that more abatement spending lowers *next* period's default threshold $\overline{m}^f_{t+1}$, which raises loan prices and dividends in the *current* period. This is reflected by the additional term $\lambda^f_t\frac{\overline{m}^f_{t+1}}{k^f_{t+1}}$ of the LHS of equation (E.54). At the same time, the expected payoff from abatement on the RHS of equation (E.54) is reduced by the factor $G(\overline{m}^f_{t+1})$, which is the expected productivity conditional on default. Put differently, firms take into account that they do not always receive the payoff from abatement investment due to the possibility of default. Equation (E.54) still implies that it is optimal to choose zero abatement if the (expected) emission tax in period $t+1$ is zero. The resource constraint becomes:

$$y_t = c_t + \frac{\theta_1(a_{t+1})^{1+\theta_2}}{1+\theta_2}k^f_{t+1} + \sum_s i^s_t\Big(1+\frac{\Psi_I}{2}\big(\frac{i^s_t}{i^s_{t-1}}-1\big)^2\Big) + \frac{\Psi_P}{2}\big(\pi_t-1\big)^2 + \sum_s \chi\varphi F(\overline{m}^s_t)l^s_t + \zeta F(\overline{\mu}_t)d_t \ .$$

The optimal condition for abatement with SLC is instead:

$$\theta_1(a_{t+1})^{\theta_2}k^f_{t+1} - \lambda^f_t\overline{m}^f_{t+1}\frac{\tau^e_{t+1}}{\widetilde{p}^f_{t+1}} - \frac{\partial q^f_t}{\partial a_{t+1}}(l^f_{t+1}-(1-\chi)\frac{l^f_t}{\pi_t}) = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\big(1-G(\overline{m}^f_{t+1})\big)\tau^e_{t+1}k^f_{t+1} \ , \quad \text{(E.55)}$$

where the loan pricing condition now explicitly depends on a_{t+1}:

$$q(\overline{m}^f_{t+1}, a_{t+1}) = \mathbb{E}_t\Big(\Big\{1-\kappa^{pen}_t + a_{t+1}(\kappa^{pen}_t - \kappa^{reward}_t)\Big\}\Xi_{t+1} + \overline{\Lambda}_{t,t+1}\Big)\mathcal{R}^f_{t+1} \ ,$$

so that the partial derivative of the bond price with respect to abatement investment is given by

$$\frac{\partial q^f_t}{\partial a_{t+1}} = \mathbb{E}_t\Big(\kappa^{pen}_t - \kappa^{reward}_t\Big)\Xi_{t+1}\mathcal{R}^f_{t+1} \ .$$

The results are shown in Panel C of Table E.1 and are overall remarkably similar to the baseline case, indicating that two additional terms that enter equation (E.55) approximately offset each other.

Climate Damages and Liquidity Valuation. In our model, climate policy trades off the benefits of emission reduction with abatement cost, taking into account adverse side effects on liquidity provision to households. To further illustrate the relevance of these side effects and the implications for optimal bank capital requirements, we provide two additional robustness checks. First, we increase the emission damage parameter to Ψ_E=0.2 such that productivity losses from unmitigated climate change amount to 8% of GDP, compared to 4% in the baseline. Panel D of Table E.1 illustrates that this substantially increases the welfare gains of climate policy, even though the effects on emissions are very similar to the baseline calibration. This applies to tilted bank capital requirements and emission taxes alike. Second, we raise the liquidity valuation parameter ω_D, such that it implies a deposit spread of 60bps, compared to 40bps in the baseline calibration. This implies a larger deposit financing wedge Ξ_t in the loan pricing condition and hence larger effects of tilted capital requirements, see Panel E of Table E.1. As we have seen in Table 3, changing Ψ_E or ω_D has only minor implications for optimal bank regulation.

E.2 Sectoral Shares and Elasticities

The sectoral structure of our economy is governed by four parameters: the substitution elasticity $\widetilde{\epsilon}$ and weighting parameter $\widetilde{\nu}$ in the CES aggregator for energy and non-energy goods (16), and the substitution elasticity ϵ and weighting parameter ν in the nested CES aggregator for clean and fossil energy (17). In this section, we discuss in more detail how changing these parameters affects our quantitative results.

Lower Energy Share $\widetilde{\nu}$. In Panel B of Table E.2, we consider a slightly lower energy share of 7% in line with the value used in Coenen et al. (2024) in a calibration to the euro area. While this does not affect the transmission of bank capital requirements on emissions, the adverse side effects on loan demand, bank balance sheets, deposit supply and welfare are less pronounced. The transmission of the SLC is broadly similar to the baseline model. However, the lower energy share implies a generally lower level of emissions such that the welfare gain of any climate policy is smaller.

Higher Energy-Non-Energy EoS $\widetilde{\epsilon}$. The elasticity between energy and non-energy goods is set to $\widetilde{\epsilon}$=0.2 in the baseline calibration, which is in line with larger multi-sector DSGE models, see for example Bartocci et al. (2024). Fried (2018) uses a value of 0.05 in a setting with endogenous innovation, while Hassler et al. (2021) estimate a value very close to zero in the short run, while stressing that the directed technological change literature usually uses values close to unity in the longer run. Since our focus is on the interplay between climate policy and bank regulation in the medium run, we also consider a higher elasticity of $\widetilde{\epsilon}$=0.4, which is used in Coenen et al. (2024), as a further robustness check. Panel C of Table E.2 demonstrates that the effect of any climate policy on emissions is larger, as the more expensive energy bundle can be substituted more easily.

	(1) No policy	(2) κ^f=100%	(3) SLC	(4) Net zero tax	(5) 5$ tax	(6) Tax shocks
Panel A: Baseline Model						
Δ Emissions	-	-0.96%	-4.59%	-100%	-18.02%	-12.01%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-59bps	-58bps	-43bps	-41bps	-41bps
Welfare Gain, CE	-	-0.13%	+0.38%	+6.93%	+1.97%	+1.26%
Panel B: Lower Energy Share $\widetilde{\nu}$						
Δ Emissions	-	-0.94%	-4.38%	-100%	-17.07%	-11.93%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.61%
Liquidity Premium	-41bps	-53bps	-52bps	-43bps	-42bps	-42bps
Welfare Gain, CE	-	-0.07%	+0.25%	+4.89%	+1.34%	+0.85%
Panel C: Higher Energy-Non-Energy EoS $\widetilde{\epsilon}$						
Δ Emissions	-	-1.11%	-4.65%	-100%	-17.38%	-12.20%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-56bps	-56bps	-43bps	-40bps	-40bps
Welfare Gain, CE	-	-0.11%	+0.36%	+6.40%	+1.88%	+1.19%
Panel D: Lower Initial Fossil Share ν						
Δ Emissions	-	-1.37%	-4.88%	-100%	-17.70%	-12.49%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-56bps	-55bps	-42bps	-41bps	-41bps
Welfare Gain, CE	-	-0.06%	+0.38%	+6.39%	+1.85%	+1.19%
Panel E: Higher Clean-Fossil EoS ϵ						
Δ Emissions	-	-1.72%	-5.27%	-100%	-18.12%	-12.88%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-59bps	-58bps	-42bps	-41bps	-41bps
Welfare Gain, CE	-	-0.03%	+0.47%	+7.40%	+2.13%	+1.38%

Table E.2: Robustness: Sectoral Shares. We compare the case of no policy (zero emission taxes and symmetric capital requirements in column 1) to different climate policies. Column (2) refers to a simple fossil penalizing capital requirement (κ^f=100%), column (3) to the sustainability linked capital requirement with κ^{pen}=100% and κ^{reward}=8%, column (4) to the emission tax consistent with net zero, column (5) to a constant 5$/tCO2 tax and column (6) to an average 5$/tCO2 tax subject to emission tax shocks. The emission change and welfare are computed relative to the baseline. Welfare is defined in equation (A.31), computed relative to the baseline without climate policies and expressed in consumption equivalents.

Lower Initial Fossil Share ν. As a third robustness check, we decrease the share of firms in the energy sector that initially use the fossil technology from 80% to 70% by setting ν=0.43 instead of ν=0.3865. In comparable three-sector models, Coenen et al. (2024) target a fossil share of 71%, based on data for the US and the EU. Nakov and Thomas (2023) use a fossil share of 75% based on global data, while Golosov et al. (2014) and Hambel and van der Ploeg (2025) consider fossil shares of around 65%. Similar to the case of a lower energy weight $\widetilde{\nu}$, emissions are generally smaller as well, which implies that all climate policies induce a smaller welfare gain than in the baseline. The lower fossil share also softens its side effects on liquidity provision, as indicated by the smaller response of the liquidity premium. Quantitatively, the results are still comparable to the baseline model (see Panel D of Table E.2).

Higher Clean-Fossil EoS ϵ. As a fourth and final robustness check, we use a higher substitution elasticity between clean and fossil energy. While we have used a substitution elasticity between clean and fossil energy of ϵ=3, in line with values used in similar models (Nakov and Thomas, 2023). In the medium run, one can also argue in favor of larger elasticities, in particular in the longer run once energy infrastructure is in place (Acemoglu et al., 2012). Hence, Panel E of Table E.2 reports the results of setting ϵ=5 instead. Unsurprisingly, the implied emission reduction and welfare gain of all climate policies is larger than in the baseline, but still of a similar magnitude.

E.3 Financial Frictions and Model Extensions

This section provides additional details to various model extensions concerning financial frictions.

Lower Intermediation Efficiency. As a first robustness check, we increase banks restructuring cost per unit of loans φ. This can be interpreted as less efficient financial intermediation. Specifically, we increase φ from its baseline value of 0.12 to 0.16, which implies that the average debt recovery rate is 70%, a value at the lower end of all OECD countries. The results in Panel A of Table E.3 indicate that our main results do not visibly depend on corporate default losses. The reason is that banks' loan pricing condition fully reflects restructuring losses so that the loan price is smaller, holding firms' risk choice constant. Consequently, firms reduce loan take-up in response.

Bond Financing. Next, we allow for the possibility of bond financing in the fossil sector. We denote the amount of bonds outstanding by b_t^f and assume that is held and priced by households, consistent with the idea that bonds are in practice held by institutional investors that are not subject to the same frictions as banks, such as mutual funds. Its pricing condition is given by:

$$q_t^{b,f} = \mathbb{E}_t \frac{\mathcal{R}_{t+1}^{b,f}}{1 + r_t^{rf}} \ , \tag{E.56}$$

where the expected payoff is discounted by the risk-free rate, such that the bond price schedule is directly related to the household sdf. Consistent with the interpretation of bonds as "arm's length debt", we assume that banks receive all revenues of a defaulting firms, but still have to pay the restructuring cost per maturing unit of loans χl_t. In addition, we assume that banks pay a monitoring cost $\widetilde{\varphi}$, which is proportional to loans outstanding. This ensures that the loan pricing condition does not strictly dominate the bond pricing condition and thereby avoids corner solutions. The *per-unit* loan and bond payoffs are given by

$$\mathcal{R}_t^{l,f} = (1-\chi) q(\overline{m}_{t+1}^f) + \frac{\chi}{\pi_t}\Big(1 - F(\overline{m}_t^f) + \frac{G(\overline{m}_t^f)}{\overline{m}_t^f} - \varphi F(\overline{m}_t^f)\Big) - \widetilde{\varphi} \ , \tag{E.57}$$

$$\mathcal{R}_t^{b,f} = (1-\chi) q(\overline{m}_{t+1}^f) + \frac{\chi}{\pi_t}\Big(1 - F(\overline{m}_t^f)\Big) \ , \tag{E.58}$$

respectively. The default threshold takes into account loans and bonds outstanding:

$$\overline{m}_{t+1}^f \equiv \frac{\chi(b_{t+1}^f + l_{t+1}^f)}{\pi_{t+1}\widetilde{p}_{t+1}^f k_{t+1}^f} \ . \tag{E.59}$$

The derivatives of the loan and bond pricing schedules with respect to $\overline{m}_t^f$ read:

$$\frac{\partial q_t^{b,f}}{\partial \overline{m}_{t+1}^f} = \mathbb{E}_t \frac{1}{1 + r_t^{rf}}\Big(- \frac{\chi F'(\overline{m}_{t+1}^f)}{\pi_{t+1}} + (1-\chi)\frac{\partial q_{t+1}^{b,f}}{\partial \overline{m}_{t+2}^f}\frac{\partial \overline{m}_{t+2}^f}{\partial \overline{m}_{t+1}^f}\Big) \ ,$$

$$\frac{\partial q_t^{l,f}}{\partial \overline{m}_{t+1}^f} = \mathbb{E}_t\Big((1-\kappa_t^f)\Xi_{t+1} + \overline{\Lambda}_{t,t+1}\Big)\cdot\Big(- \frac{\chi G(\overline{m}_{t+1}^f)}{\pi_{t+1}(\overline{m}_{t+1}^f)^2} - \frac{\chi\varphi F'(\overline{m}_{t+1}^f)}{\pi_{t+1}} + (1-\chi)\frac{\partial q_{t+1}^{l,f}}{\partial \overline{m}_{t+2}^f}\frac{\partial \overline{m}_{t+2}^f}{\partial \overline{m}_{t+1}^f}\Big) \ .$$

As the composition of bank and bond debt is not constant over the business cycle, this would introduce excessive fluctuations in bank balance sheets and the deposit-to-GDP ratio, which is a target moment in our calibration. To ensure a fair comparison between this extension and the baseline model, we keep the liquidity curvature parameter γ_D at its baseline value and introduce an additional quadratic cost that firms incur when they deviate from the steady state bond-to-loan ratio $b_t/l_t = 1$. The firm maximization problem is now given by

$$\begin{aligned}
\max_{b^f_{t+1},k^f_{t+1},l^f_{t+1},\overline{m}^f_{t+1}} \quad & -(1+\tau^f)\psi^f_t k^f_{t+1} - \frac{\Psi_B}{2}(b^f_{t+1} - l^f_{t+1})^2 \\
& + q^b(\overline{m}^f_{t+1})\Big(b^f_{t+1} - (1-\chi)\frac{b^f_t}{\pi_t}\Big) + q^l(\overline{m}^f_{t+1})\Big(l^f_{t+1} - (1-\chi)\frac{l^f_t}{\pi_t}\Big) \\
& + \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Bigg(\int_{\overline{m}^f_{t+1}}^{\infty} \widetilde{p}^f_{t+1} m k^f_{t+1} - \chi\frac{b^f_{t+1} + l^f_{t+1}}{\pi_{t+1}} dF(m) + (1+\tau^f)\psi^f_{t+1}(1-\delta_K)k^f_{t+1} \\
& \qquad + q^b(\overline{m}^f_{t+2})\Big(b^f_{t+2} - (1-\chi)\frac{b^f_{t+1}}{\pi_{t+1}}\Big) + q^l(\overline{m}^f_{t+2})\Big(l^f_{t+2} - (1-\chi)\frac{l^f_{t+1}}{\pi_{t+1}}\Big)\Bigg) ,
\end{aligned}$$

subject to the default threshold (E.59), the bond price schedule and the debt price schedule. The first-order condition for bonds and loans are given by

$$\begin{aligned}
& q^b(\overline{m}^f_{t+1}) - \Psi_b(b^f_{t+1} - l^f_{t+1}) - \lambda^f_t \frac{\overline{m}^f_{t+1}}{b^f_{t+1} + l^f_{t+1}} \\
& \quad = \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}}\Big(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^b(\overline{m}^f_{t+2})\Big) ,
\end{aligned} \tag{E.60}$$

and

$$\begin{aligned}
& q^l(\overline{m}^f_{t+1}) + \Psi_b(b^f_{t+1} - l^f_{t+1}) - \lambda^f_t \frac{\overline{m}^f_{t+1}}{b^f_{t+1} + l^f_{t+1}} \\
& \quad = \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}}\Big(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^l(\overline{m}^f_{t+2})\Big) .
\end{aligned} \tag{E.61}$$

As in the baseline model, optimal capital satisfies

$$\psi^f_t(1+\tau^f) - \lambda^f_t \frac{\overline{m}^f_{t+1}}{k^f_{t+1}} = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\psi^f_{t+1}(1-\delta_K)(1+\tau^f) + \widetilde{p}^f_{t+1}\big(1 - G(\overline{m}^f_{t+1})\big)\Big) , \tag{E.62}$$

whereas the risk choice $\overline{m}^f_{t+1}$ is determined according to

$$-\lambda^f_t - \frac{\partial q^b_t}{\partial \overline{m}^f_{t+1}}\Big(b^f_{t+1} - (1-\chi)\frac{b^f_t}{\pi_t}\Big) - \frac{\partial q^l_t}{\partial \overline{m}^f_{t+1}}\Big(l^f_{t+1} - (1-\chi)\frac{l^f_t}{\pi_t}\Big) \tag{E.63}$$
$$= \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\bigg(\Big(b^f_{t+2} - (1-\chi)\frac{b^f_{t+1}}{\pi_{t+1}}\Big)\frac{\partial q^b_{t+1}}{\partial \overline{m}^f_{t+2}} + \Big(l^f_{t+2} - (1-\chi)\frac{l^f_{t+1}}{\pi_{t+1}}\Big)\frac{\partial q^l_{t+1}}{\partial \overline{m}^f_{t+2}}\bigg)\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}} \ .$$

To pin down the policy function for the default threshold $\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}}$, the possibility of issuing bonds has to be taken into account. As before, we re-arrange the first-order condition for loans, equation (E.61), for λ^f_t:

$$\lambda^f_t = \frac{b^f_{t+1} + l^f_{t+1}}{\overline{m}^f_{t+1}}\Big(q^l(\overline{m}^f_{t+1}) + \Psi_b(b^f_{t+1} - l^f_{t+1})\Big)$$
$$- \mathbb{E}_t\frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}}\frac{b^f_{t+1} + l^f_{t+1}}{\overline{m}^f_{t+1}}\bigg(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^l(\overline{m}^f_{t+2})\bigg) \ ,$$

and plug this expression into the first-order condition for capital (E.62) to get

$$(1+\tau^f)\psi^f_t - \mathbb{E}_t\frac{b^f_{t+1} + l^f_{t+1}}{\pi_{t+1}k^f_{t+1}}\left(\Big(q^l(\overline{m}^f_{t+1}) + \Psi_b(b^f_{t+1} - l^f_{t+1})\Big)\pi_{t+1} - \widetilde{\Lambda}_{t,t+1}\Big\{\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^l(\overline{m}^f_{t+2})\Big\}\right)$$
$$= \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\bigg((1-\delta_k)\psi^f_{t+1}(1+\tau^f) + (1 - G(\overline{m}^f_{t+1}))\widetilde{p}^f_{t+1}\bigg) \ .$$

Multiplying both sides by $\frac{\chi}{\widetilde{p}^f_{t+1}}$ yields:

$$\frac{\chi}{\widetilde{p}^f_{t+1}}\psi^f_t(1+\tau^f) - \overline{m}^f_{t+1}\mathbb{E}_t\left(\Big(q^l(\overline{m}^f_{t+1}) + \Psi_b(b^f_{t+1} - l^f_{t+1})\Big)\pi_{t+1} - \widetilde{\Lambda}_{t,t+1}\Big\{\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^l(\overline{m}^f_{t+2})\Big\}\right)$$
$$= \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\bigg(\frac{\chi}{\widetilde{p}^f_{t+1}}(1-\delta_k)\psi^f_{t+1}(1+\tau^f) + \chi(1 - G(\overline{m}^f_{t+1}))\bigg) \ .$$

Differentiating this expression w.r.t. $\overline{m}^f_{t+1}$, we obtain

$$-\Big(q^l(\overline{m}^f_{t+1}) + \Psi_b(b^f_{t+1} - l^f_{t+1})\Big)\pi_{t+1} + \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi)q^l(\overline{m}^f_{t+2})\Big)$$
$$- \overline{m}^f_{t+1}\mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\left(\frac{\partial q^{l,f}_t}{\partial \overline{m}^f_{t+1}}\pi_{t+1} + \chi f(\overline{m}^f_{t+1}) + (1-\chi)\frac{\partial q^{l,f}_{t+1}}{\partial \overline{m}^f_{t+2}}\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}}\right) = -\chi\mathbb{E}_t\widetilde{\Lambda}_{t,t+1}g(\overline{m}^f_{t+1}) \ .$$

As before, we exploit the relationship $g(\overline{m}^f_{t+1})$ satisfies $g(\overline{m}^f_{t+1}) = \overline{m}^f_{t+1} f(\overline{m}^f_{t+1})$ to obtain the following additional equilibrium condition:

$$q^l(\overline{m}^f_{t+1}) + \overline{m}^f_{t+1} \frac{\partial q^{l,f}_t}{\partial \overline{m}^f_{t+1}} + \Psi_b(b^f_{t+1} - l^f_{t+1})$$
$$= \mathbb{E}_t \frac{\widetilde{\Lambda}_{t,t+1}}{\pi_{t+1}} \Big(\chi(1 - F(\overline{m}^f_{t+1})) + (1-\chi) \Big\{ q^l(\overline{m}^f_{t+2}) + \overline{m}^f_{t+1} \frac{\partial q^{l,f}_{t+1}}{\partial \overline{m}^f_{t+2}} \frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}^f_{t+1}} \Big\} \Big) \,. \tag{E.64}$$

The US loan share is around one third (Crouzet, 2017 and Dempsey, 2024), while in the euro area loans make up around two thirds of corporate sector borrowing (Darmouni and Papoutsi, 2025). Hence, we calibrate $\widetilde{\varphi}$=0.0055 in order to match a bond share of 50%, striking a middle ground between the US and the euro area. The parameter governing the cost of deviating from the bond-to-loan ratio is set to Ψ_B=0.15, which ensures that the excess volatility of deposits over GDP matches its target.

Sector-Specific Banks. Third, we drop the assumption that banks are perfectly diversified across sectors. The 2022 ECB Banking Supervision climate stress test has revealed a substantial heterogeneity of banks' exposure to the fossil energy sector. This issue has also been recognized by the EBA (2022), Chapter 8, which discusses highly concentrated bank portfolios in the context of environmental risks. As the degree of imperfect diversification of bank portfolios across sectors is difficult to calibrate in our setting, we assume that banks are only active in one sector, which provides an upper bound on limited bank diversification across sectors.

Each (representative) sector-specific bank extends deposits d^s_t to households. For simplicity, we assume that deposits of each bank are perfectly substitutable. Thus, only aggregate deposits d_t are relevant for households' valuation of liquidity services and the deposit market clearing condition becomes $d_t = d^c_t + d^f_t + d^n_t$. The bank failure thresholds are sector specific and given by $\overline{\mu}^s_t = \frac{\mathcal{R}^s_t l^s_t}{(1+i^D_{t-1})d^s_t}$ for $s \in \{c, f, n\}$. Solving the profit maximization problem yields the following loan pricing condition:

$$q^s_t = \mathbb{E}_t \Big((1-\kappa^s_t) \Big\{ \frac{1}{1+r^D_t} - \Lambda_{t,t+1}\big(1 - F(\overline{\mu}^s_{t+1})\big) \Big\} + \Lambda_{t,t+1}\big(1 - G(\overline{\mu}^s_{t+1})\big) \Big) \mathcal{R}^s_{t+1} \,. \tag{E.65}$$

Different to the baseline case (5), the bank failure probability is sector-specific, such that

	(1) Baseline	(2) κ^f=100%	(3) SLC	(4) Net zero tax	(5) 10$ tax	(6) Tax shocks
Panel A: Baseline Model						
Δ Emissions	-	-0.96%	-4.59%	-100%	-18.02%	-12.01%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-40bps	-59bps	-58bps	-43bps	-41bps	-41bps
Welfare Gain, CE	-	-0.13%	+0.38%	+6.93%	+1.97%	+1.26%
Panel B: High Restructuring Cost						
Δ Emissions	-	-0.98%	-4.65%	-100%	-17.15%	-12.01%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-42bps	-62bps	-61bps	-45bps	-43bps	-43bps
Welfare Gain, CE	-	-0.14%	+0.38%	+6.95%	+1.99%	+1.26%
Panel C: Bond Financing						
Δ Emissions	-	-0.59%	-0.60%	-100%	-17.18%	-12.04%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.63%
Liquidity Premium	-41bps	-50bps	-50bps	-42bps	-41bps	-41bps
Welfare Gain, CE	-	+0.06%	+0.07%	+6.45%	+1.88%	+1.18%
Panel D: Heterogeneous Banks						
Δ Emissions	-	-0.95%	-4.61%	-100%	-17.15%	-12.01%
Bank Failure Rate	0.59%	0.59%	0.59%	0.59%	0.59%	0.63%
Liquidity Premium	-41bps	-59bps	-58bps	-43bps	-41bps	-41bps
Welfare Gain, CE	-	-0.14%	+0.37%	+6.96%	+1.99%	+1.26%
Panel E: Risk-Free Reserve Asset						
Δ Emissions	-	-1.13%	-4.28%	-100%	-17.16%	-12.01%
Bank Failure Rate	0.59%	<0.1%	<0.1%	0.59%	0.59%	0.62%
Liquidity Premium	-40bps	-47bps	-42bps	-42bps	-41bps	-41bps
Welfare Gain, CE	-	-0.20%	+0.25%	+6.94%	+1.99%	+1.25%

Table E.3: Robustness: Financial Frictions and Extensions. We compare the case of no policy (zero emission taxes and symmetric capital requirements in column 1) to different climate policies. Column (2) refers to a simple fossil penalizing capital requirement (κ^f=100%), column (3) to the sustainability linked capital requirement with κ^{pen}=100% and κ^{reward}=8%, column (4) to the emission tax consistent with net zero, column (5) to a constant 5$/tCO2 tax and column (6) to an average 5$/tCO2 tax subject to emission tax shocks. The emission change and welfare are computed relative to the baseline. Welfare is defined in equation (A.31), computed relative to the baseline without climate policies and expressed in consumption equivalents.

both the deposit financing wedge Ξ^s_{t+1} and the bank-owner sdf $\overline{\Lambda}^s_{t,t+1}$ are sector-specific. The aggregate bank failure rate is defined as

$$F(\overline{\mu}_{t+1}) \equiv \frac{d^c_t F(\overline{\mu}^c_{t+1}) + d^f_t F(\overline{\mu}^f_{t+1}) + d^n_t F(\overline{\mu}^n_{t+1})}{d_t} \,. \tag{E.66}$$

Weighting sector-specific bank failure rates by deposits is consistent with the assumption that bailout losses incurred by the fiscal authority are proportional to the quantity of deposits that are bailed out. Panel D of Table E.3 summarizes the results. Again, the effects of tilted bank capital requirements on aggregate emissions are very similar to the baseline. The transmission of tilted bank regulation and the adverse effects of climate policies on liquidity provision is quite similar to the baseline model. In sharp contrast to the baseline model, however, the bank failure rate is essentially independent of the climate policy regime, which simply follows from the very low weight on $F(\overline{\mu}^f_{t+1})$ in equation (E.66). This gives rise to the slightly smaller optimal capital requirement under net zero taxes that we reported in Table 3.

Risk-Free Reserve Assets. Fourth and finally, we extend the model by introducing nominally risk-free assets, i.e. government bonds. Importantly, we assume that the bank-specific risk shock also applies to government bond holdings so that the realized payoff from the portfolio is given by $\mu_t\big(\sum_s \mathcal{R}^s_t l^s_t + \mathcal{R}^g_t g_t\big)$. The pricing condition q^g_t for government bonds g_t is completely analogous to the loan pricing condition:

$$q^g_t = \mathbb{E}_t\Big((1-\kappa^g_t)\Big\{\frac{1}{1+r^D_t} - \Lambda_{t,t+1}\big(1-F(\overline{\mu}_{t+1})\big)\Big\} + \Lambda_{t,t+1}\big(1-G(\overline{\mu}_{t+1})\big)\Big)\mathcal{R}^g_{t+1}\,. \tag{E.67}$$

Different from the loan pricing condition, equation (E.67), the government bond payoff merely reflects its random-maturity structure but does not contain terms related to default risk:

$$\mathcal{R}^g_t = \chi + (1-\chi)q^g_t\,. \tag{E.68}$$

Government bonds are supplied inelastically and we set g_t=0.6, matching a government debt-to-GDP ratio of 100%. We recalibrate the weighting parameter on households' valuation of liquidity services to ω_D=0.03 as deposits are much less scarce in the presence of government bonds.

Panel E of Table E.3 shows that the effect of tilted bank capital regulation is comparable to the baseline model, but its adverse side effects on liquidity provision are less pronounced since fossil loans make up a smaller share of the aggregate bank balance sheet. However, these side effects receive a larger welfare weight due to the higher value of ω_D. Hence, the simple fossil capital requirement implies a larger welfare loss than in the baseline and the welfare gain of a SLC is smaller. By contrast, the welfare gain of net zero emission taxes and the welfare losses of climate policy uncertainty are hardly affected by the presence of nominally risk-free assets.

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